

2.4 Zeros and singularities

2.4.1 Zeros

$f \not\equiv 0$ holomorphic, $f(a) = 0 \Rightarrow f(z) = (z - a)^m g(z)$, $g(a) \neq 0$.
Zeros are isolated unless $f \equiv 0$.

Identity theorem: $f = g$ on a set with a limit point in $U \Rightarrow f \equiv g$ on U .

2.4.2 Classification of isolated singularities

- *Removable:* Laurent series has no negative terms; $\lim_{z \rightarrow a} f(z)$ finite.
- *Pole:* finite negative part; $\lim_{z \rightarrow a} f(z) = \infty$.
- *Essential:* infinitely many negative terms; $\lim_{z \rightarrow a} f(z)$ does not exist.

HLTiOM

Pole criterion

If $|f(z)| \rightarrow \infty$ as $z \rightarrow a$, then a is a pole. a is a pole $\iff 1/f$ has a removable zero.

2.5 Residue theorem

2.5.1 Residue

$\text{Res}(f, a) = c_{-1}$ for a Laurent expansion at a .

- Simple pole: $\text{Res}(f, a) = \lim_{z \rightarrow a} (z - a)f(z) = \frac{g(a)}{h'(a)}$ if $f = g/h$, $h(a) = 0$, $h'(a) \neq 0$.
- Pole of order m : $\text{Res}(f, a) = \frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} ((z-a)^m f(z))$.

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2.2.2 Cauchy's integral formula

$f \in H(U)$, $\overline{\Delta_r(a)} \subset U$. For $z \in \Delta_r(a)$:

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta-a|=r} \frac{f(\zeta)}{\zeta-z} d\zeta.$$

Consequence (derivatives): $f^{(n)}(a) = \frac{n!}{2\pi i} \oint \frac{f(\zeta)}{(\zeta-a)^{n+1}} d\zeta$.

2.2.3 Cauchy estimate and Liouville

$|f^{(n)}(a)| \leq \frac{n!}{r^n} \max_{|\zeta-a|=r} |f(\zeta)|$. *Liouville*: a bounded entire function is constant. *Fundamental theorem of algebra*: every non-constant polynomial has a root in $\mathbb{C}$.

2.3 Power series and analyticity

2.3.1 Taylor series

Holomorphic f is analytic: $f(z) = \sum_{n \geq 0} a_n(z-a)^n$, converging in $\Delta_R(a)$ where R = distance to the nearest singularity. $a_n = f^{(n)}(a)/n!$.

2.3.2 Laurent series

f holomorphic on an annulus $r < |z-a| < R$:

$$f(z) = \sum_{n=-\infty}^{\infty} c_n(z-a)^n, \quad c_n = \frac{1}{2\pi i} \oint_{|\zeta-a|=\rho} \frac{f(\zeta)}{(\zeta-a)^{n+1}} d\zeta.$$

2.1.2 Polar form

In polar coordinates $z = re^{i\theta}$, write $f = u(r, \theta) + iv(r, \theta)$. The CR equations become

$$u_r = \frac{1}{r} v_\theta, \quad v_r = -\frac{1}{r} u_\theta.$$

2.1.3 Harmonic functions

Holomorphic $f = u + iv \Rightarrow \Delta u = u_{xx} + u_{yy} = 0$, $\Delta v = 0$; u and v are harmonic. On a simply connected domain, every harmonic u admits a harmonic conjugate v such that $u + iv$ is holomorphic:

$$v(x, y) = \int_{(x_0, y_0)}^{(x, y)} (-u_y dx + u_x dy).$$

2.1.4 Wirtinger calculus

$$\partial_z = \frac{1}{2}(\partial_x - i\partial_y), \quad \partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y).$$

f is holomorphic $\iff \partial_{\bar{z}} f = 0$; then $f'(z) = \partial_z f$. Also $\Delta = 4\partial_z \partial_{\bar{z}}$.

2.2 Cauchy's integral theorem and formula

2.2.1 Cauchy's theorem

$f \in H(U)$, γ is a null-homotopic closed curve in $U \Rightarrow \oint_\gamma f(z) dz = 0$.

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Chapter 2

Complex Analysis

Notation. $U \subset \mathbb{C}$ open; $\Delta_r(a) = \{z : |z - a| < r\}$; ∂U — boundary. $H(U)$ — holomorphic (complex differentiable) functions on U ; $f^{(n)}$ — n -th derivative. $\oint_\gamma f dz$ — contour integral; $\text{ind}_\gamma(a) = \frac{1}{2\pi i} \oint_\gamma \frac{dz}{z-a}$ — winding number. $\text{Res}(f, a)$ — residue at a . A curve γ is *null-homotopic* in U if it can be continuously contracted to a point within U . An *entire* function is holomorphic on all of $\mathbb{C}$. An *annulus* is a region $r < |z - a| < R$.

2.1 Cauchy–Riemann equations

2.1.1 Cartesian form

$f = u + iv$ is holomorphic on $U \iff u, v \in C^1$ and

$$u_x = v_y, \quad u_y = -v_x.$$

Equivalently $\partial_{\bar{z}} f = 0$.

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1.9.6 Riemann rearrangement theorem

A conditionally convergent series can be rearranged so that its sum becomes any $L \in \mathbb{R} \cup \{\pm\infty\}$.

1.9.7 Cauchy product and Mertens’ theorem

If $\sum a_n$ converges absolutely to A and $\sum b_n$ converges (conditionally) to B , then the Cauchy product $c_n = \sum_{k=0}^n a_k b_{n-k}$ yields $\sum c_n = AB$.

1.9.2 Abel test

Statement. Let $\sum a_n$ converge and $\{b_n\}$ be monotone and bounded. Then $\sum a_n b_n$ converges.

When to use. Multiplying a convergent series by an oscillating factor; power series at the boundary of the convergence disk.

1.9.3 Cauchy condensation test

$a_n \geq 0$ monotonically non-increasing $\Rightarrow \sum a_n$ and $\sum 2^k a_{2^k}$ converge or diverge together. Application: Bertrand series $\sum 1/(n \ln^p n)$.

1.9.4 Integral test and estimate

For $f \geq 0$ decreasing on $[1, \infty)$:

$$\int_1^{n+1} f(x) dx \leq \sum_{k=1}^n f(k) \leq f(1) + \int_1^n f(x) dx.$$

In the limit: $\sum_{n=1}^{\infty} f(n)$ converges $\iff \int_1^{\infty} f(x) dx$ converges.

When to use. Estimating partial sums of slowly decaying series (e.g. $\sum 1/k^s$, $\sum 1/(k \ln k)$). The integral bounds give O and Θ asymptotics.

1.9.5 Kummer, Raabe, Gauss tests

Kummer. $a_n > 0$, $p_n > 0$, $K_n = p_n \frac{a_n}{a_{n+1}} - p_{n+1}$.

- $\liminf K_n > 0 \Rightarrow$ convergence.
- $\limsup K_n < 0$ and $\sum 1/p_n = \infty \Rightarrow$ divergence.

Raabe. $R_n = n(\frac{a_n}{a_{n+1}} - 1)$. Convergence when $\liminf R_n > 1$.

Gauss. $\frac{a_n}{a_{n+1}} = 1 + \frac{\mu}{n} + O(n^{-1-\delta}) \Rightarrow$ convergence when $\mu > 1$.

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When to use. Showing that an equation defines a C^1 curve/surface near a non-degenerate point; computing partial derivatives of implicitly defined functions.

1.8.3 Lagrange multipliers

Extremum of $f(x)$ under $g_1(x) = \cdots = g_k(x) = 0$: $\nabla f = \sum \lambda_i \nabla g_i$ at a regular point. The k constraints reduce dimension; λ_i are Lagrange multipliers.

When to use. Optimization with equality constraints; inequalities by extremizing a function over a sphere, simplex, etc.

1.8.4 Change of variables in multiple integrals

$x = \Phi(u)$, Φ is a C^1 diffeomorphism:

$$\int_{\Phi(\Omega)} f(x) dx = \int_{\Omega} f(\Phi(u)) |\det J_{\Phi}(u)| du.$$

Polar in $\mathbb{R}^2$: $dx dy = r dr d\theta$. *Spherical in $\mathbb{R}^3$:* $dx dy dz = r^2 \sin \theta dr d\theta d\phi$.

1.9 Sequences and series

1.9.1 Dirichlet test

Statement. If the partial sums $A_n = \sum_{k=1}^n a_k$ are bounded and b_n decreases monotonically to 0, then $\sum a_n b_n$ converges.

When to use. Alternating series (Leibniz is a special case), series $\sum (\sin n\theta) b_n$, $\sum (\cos n\theta) b_n$, $\sum z^n b_n$ ($|z| = 1$, $z \neq 1$) with $b_n \downarrow 0$.

Remainder estimate (Leibniz). For $\sum (-1)^n b_n$ with $b_n \downarrow 0$: $|R_n| \leq b_{n+1}$. If $b_n = f(n)$ with $f \in C^1$, then by MVT (1.2.1) $b_n - b_{n+1} = f'(\xi_n)$, $\xi_n \in (n, n+1)$ — for more precise decay estimates.

Idea. Abel transformation: $\sum_{k=1}^N a_k b_k = A_N b_N - \sum_{k=1}^{N-1} A_k (b_{k+1} - b_k)$.

1.7.3 Euler's reflection formula

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x} \text{ for } x \notin \mathbb{Z}.$$

1.7.4 Frullani integral

f is continuous, $A = f(0^+)$, $B = \lim_{x \rightarrow \infty} f(x)$. Then for $a, b > 0$:

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = (A - B) \ln \frac{b}{a}.$$

1.7.5 Differentiation under the integral

$$\frac{d}{dt} \int_{a(t)}^{b(t)} f(x, t) dx = f(b(t), t) b'(t) - f(a(t), t) a'(t) + \int_{a(t)}^{b(t)} \partial_t f(x, t) dx.$$

1.7.6 Symmetry $x \mapsto a + b - x$ and inversion $x \mapsto 1/x$

Symmetry $x \mapsto a + b - x$: $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$. If $f(x) + f(a + b - x) = h(x)$, the integral equals $\frac{1}{2} \int_a^b h$.
Inversion $x \mapsto 1/x$: the measure dx/x is invariant. $\int_0^\infty f(x)/x dx = \int_0^\infty f(1/x)/x dx$.

1.8 Multivariable calculus

1.8.1 Gradient, Jacobian, directional derivative

For $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $J_F = (\partial_j F_i)$. $D_v F(a) = J_F(a)v$. Chain rule: $J_{F \circ G} = J_F(G) \cdot J_G$.

1.8.2 Implicit function theorem

$F : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$ is C^1 , $F(a, b) = 0$, $\det \partial_y F(a, b) \neq 0$. Then near a there exists a unique C^1 function $g(x)$ with $F(x, g(x)) = 0$ and $J_g = -(\partial_y F)^{-1} \partial_x F$.

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1.6.9 Standard substitutions and tricks

The substitution $z = \varphi(y)$ simplifies the equation if y' can be expressed via z' :

- $f'/f = (\ln|f|)'$: if the equation contains $\frac{y'}{y}$, set $z = \ln|y|$.
- $ff' = \left(\frac{f^2}{2}\right)'$: if it contains yy' , set $z = y^2$.
- $f^n f' = \left(\frac{f^{n+1}}{n+1}\right)'$: contains $y^n y'$, set $z = y^{n+1}$.
- $f'/f^2 = \left(-\frac{1}{f}\right)'$: set $z = 1/y$ for y'/y^2 .
- $f'/f^n = \left(\frac{1}{(1-n)f^{n-1}}\right)'$, $n \neq 1$: set $z = y^{1-n}$.
- $f'/\sqrt{f} = (2\sqrt{f})'$.

1.7 Gamma function, beta function, special integrals

1.7.1 Gamma function

$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$ ($\operatorname{Re} s > 0$). $\Gamma(s+1) = s\Gamma(s)$. $\Gamma(n+1) = n!$, $\Gamma(1/2) = \sqrt{\pi}$.

1.7.2 Beta function

$B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$.

Trigonometric form: $B(p, q) = 2 \int_0^{\pi/2} \sin^{2p-1} \theta \cos^{2q-1} \theta d\theta$.

1.6.3 Bernoulli equation

Statement. $y' + p(x)y = q(x)y^n$. Substitution $t = y^{1-n}$ reduces to a linear equation.

1.6.4 Riccati equation

Statement. $y' = a(x)y^2 + b(x)y + c(x)$. If a particular solution y_1 is known, the substitution $y = y_1 + 1/t$ reduces to a linear equation.

1.6.5 Exact equations

Statement. $M(x, y)dx + N(x, y)dy = 0$, $\partial_y M = \partial_x N$. Then $F(x, y) = C$, where $\partial_x F = M$, $\partial_y F = N$.

Integrating factor: if $(M_y - N_x)/N$ depends only on x , then $\mu = e^{\int \frac{M_y - N_x}{N} dx}$.

1.6.6 Homogeneous equations

Statement. $y' = f(y/x)$. Substitution $t = y/x$ reduces to separable: $t + xt' = f(t)$.

1.6.7 Linear equations with constant coefficients

Statement. $a_n y^{(n)} + \dots + a_0 y = f(x)$.

- Homogeneous: characteristic equation $a_n \lambda^n + \dots + a_0 = 0$. A root λ of multiplicity m gives solutions $x^k e^{\lambda x}$, $k = 0, \dots, m-1$.
- Particular solution: method of undetermined coefficients (for $f = P(x)e^{\alpha x} \cos \beta x$, etc.) or variation of parameters.

1.6.8 Euler equation

Statement. $a_n x^n y^{(n)} + \dots + a_0 y = f(x)$. Substitution $x = e^t$ reduces to an equation with constant coefficients.

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1.5 Banach fixed-point theorem

Statement. X is complete, $T : X \rightarrow X$, $d(Tx, Ty) \leq k d(x, y)$ with $k < 1$. Then: unique fixed point x^* , iterates $x_{n+1} = Tx_n$ converge with $d(x_n, x^*) \leq \frac{k^n}{1-k} d(x_1, x_0)$.

1.5.1 Contraction in power

Statement. If T^n is a contraction for some $n \geq 1$, then T has a unique fixed point (T itself need not be a contraction).

1.5.2 Picard–Lindelöf theorem

Statement. $y' = f(x, y)$, $y(x_0) = y_0$, f continuous and locally Lipschitz in y . The Picard operator $(Ty)(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt$ is a contraction on $C[x_0 - h, x_0 + h]$ for small h . Hence a unique local solution exists.

When to use. Existence and uniqueness of solutions to ODEs; iterative construction (Picard iteration) with explicit error estimates via the Banach theorem (1.5).

1.6 Ordinary differential equations

1.6.1 Separable equations

Statement. $y' = f(x)g(y)$. Solution: $\int \frac{dy}{g(y)} = \int f(x) dx$.

1.6.2 First-order linear equations

Statement. $y' + p(x)y = q(x)$. Solution: $y = e^{-\int p} (C + \int q e^{\int p})$ (variation of constants or integrating factor $e^{\int p}$).

1.4.2 Related Cauchy-type equations

Equation	Solution
$f(x+y) = f(x)f(y)$	$f \equiv 0$ or $f(x) = e^{cx}$
$f(xy) = f(x) + f(y)$ on $\mathbb{R}_{>0}$	$f(x) = c \ln x$
$f(xy) = f(x)f(y)$ on $\mathbb{R}_{>0}$, $f > 0$	$f(x) = x^c$

Method. Reduce to Cauchy via $\ln$ or $\exp$.

1.4.3 Jensen's equation

Claim. $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2}$. Under continuity: $f(x) = ax + b$.

1.4.4 D'Alembert's equation (cosine)

Claim. $f(x+y) + f(x-y) = 2f(x)f(y)$, $f \not\equiv 0$. Under continuity: $f(x) = \cos(\alpha x)$ or $f(x) = \cosh(\alpha x)$. $f(0) = 1$.

1.4.5 Quadratic equation

Claim. $f(x+y) + f(x-y) = 2f(x) + 2f(y)$. Under continuity: $f(x) = cx^2$.

1.4.6 Techniques: substitutions, shift, contraction

Substitutions. $x = y = 0$ (find $f(0)$), $y = x$ (relate $f(2x)$ to $f(x)$), $y = -x$ (parity), symmetry $P(x, y)$ and $P(y, x)$.

Shift amplification. From $f(x+a) \leq f(x) - c$ iteration gives $f(x+na) \rightarrow -\infty$ — contradiction with $f \geq 0$ or boundedness.

Contraction. $|f(x) - f(y)| \leq c \cdot q^n$ for all $n \geq 1$ with $q < 1 \Rightarrow f(x) = f(y)$ (send $n \rightarrow \infty$).

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Power mean. $M_r(x; w) = (\sum w_i x_i^r)^{1/r}$ is strictly increasing in r (unless all x_i are equal). $M_0 = \prod x_i^{w_i}$.
Idea. From Young's inequality $ab \leq a^p/p + b^q/q$ after normalization.

1.3.5 Young and Minkowski

Young: $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$ for $1/p + 1/q = 1$, $a, b \geq 0$. ε -version:
 $ab \leq \varepsilon a^p/p + \varepsilon^{-q/p} b^q/q$.
Minkowski: $(\sum (a_i + b_i)^p)^{1/p} \leq (\sum a_i^p)^{1/p} + (\sum b_i^p)^{1/p}$ for $p \geq 1$.
Integral form: $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

1.3.6 Rearrangement and Chebyshev

Statement. For $a_1 \leq \dots \leq a_n$, $b_1 \leq \dots \leq b_n$ and a permutation σ :

$$\sum a_i b_{n+1-i} \leq \sum a_i b_{\sigma(i)} \leq \sum a_i b_i.$$

Chebyshev: $n \sum a_i b_i \geq (\sum a_i)(\sum b_i)$ for similarly ordered sequences.
Integral form: $(b-a) \int_a^b f g \geq \int_a^b f \int_a^b g$ for similarly monotone f, g .
Idea. Identity $n \sum a_i b_i - \sum a_i \sum b_i = \frac{1}{2} \sum_{i,j} (a_i - a_j)(b_i - b_j)$.

1.4 Functional equations on $\mathbb{R}$

1.4.1 Cauchy's equation (additive)

Claim. $f(x + y) = f(x) + f(y)$. Under any regularity condition (continuity at a point, monotonicity, boundedness on a set of positive measure): $f(x) = cx$, $c = f(1)$.

Without regularity. Discontinuous additive functions exist (via a Hamel basis). Their graph is dense in $\mathbb{R}^2$.

1.3 Inequalities

1.3.1 AM–GM

Statement. $\frac{x_1 + \dots + x_n}{n} \geq \sqrt[n]{x_1 \cdots x_n}$ for $x_i \geq 0$. Equality when $x_1 = \dots = x_n$.

Weighted: $\sum w_i x_i \geq \prod x_i^{w_i}$ with $\sum w_i = 1$.

Idea. Take logs + Jensen for $\ln$ (concave). Or Cauchy's forward-backward induction.

1.3.2 Cauchy–Schwarz

Statement. $(\sum a_i b_i)^2 \leq (\sum a_i^2)(\sum b_i^2)$. Integral form: $(\int fg)^2 \leq \int f^2 \int g^2$. Equality when $a_i = \lambda b_i$ (see also 3.6.5).

Engel form (Titu): $\sum \frac{a_i^2}{b_i} \geq \frac{(\sum a_i)^2}{\sum b_i}$ for $b_i > 0$.

Idea. The quadratic polynomial $\sum (a_i + tb_i)^2 \geq 0$ has non-negative discriminant.

1.3.3 Jensen

Statement. For convex f : $f(\sum w_i x_i) \leq \sum w_i f(x_i)$ with $\sum w_i = 1$. For concave — reverse. Integral form: $f(\int x d\mu) \leq \int f d\mu$.

When to use. Any convex/concave function in an inequality.

Characterization of convexity. f is convex on $I \iff f'' \geq 0$ (if C^2), or secants are monotone, or there is a supporting line at each point. A convex function on an interval is continuous in the interior and has one-sided derivatives.

Hermite–Hadamard: For convex f on $[a, b]$: $f(\frac{a+b}{2}) \leq \frac{1}{b-a} \int_a^b f \leq \frac{f(a)+f(b)}{2}$.

1.3.4 Hölder and power mean

Statement (Hölder). For $1/p + 1/q = 1$: $\sum |a_i b_i| \leq (\sum |a_i|^p)^{1/p} (\sum |b_i|^q)^{1/q}$. Integral form: $\|fg\|_1 \leq \|f\|_p \|g\|_q$.

Chapter 1

Real Analysis

Notation. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ — usual. $C[a, b]$ — continuous functions on $[a, b]$; C^k — k times differentiable with continuous k -th derivative. $f^{(n)}$ — n -th derivative; $f^{(0)} = f$. $\sum_{n=1}^{\infty} a_n$ — series; $\limsup, \liminf$. $\Gamma(s)$ — Gamma function; $B(p, q)$ — Beta function. $O(\cdot), o(\cdot), \sim$ — asymptotic notation.

1.1 Limits of sequences and functions

1.1.1 Stolz–Cesàro theorem

Statement. b_n strictly increases to $+\infty$ (or $a_n, b_n \rightarrow 0$, b_n strictly decreases). If $\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = L$, then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$.

When to use. Limits of the form $\frac{\infty}{\infty}$ or $\frac{0}{0}$ involving differences. Computing $\lim \frac{\ln n}{n}, \lim \frac{n}{\sqrt[n]{n!}}$.

Corollaries. Cesàro mean: $a_n \rightarrow L \Rightarrow \frac{1}{n} \sum_{k \leq n} a_k \rightarrow L$. Geometric: $a_{n+1}/a_n \rightarrow L \Rightarrow \sqrt[n]{a_n} \rightarrow L$.

1.1.2 Limsup and liminf

$$\limsup_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k, \quad \liminf_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} a_k.$$

- $\limsup(a_n + b_n) \leq \limsup a_n + \limsup b_n$;
- $\limsup(a_n b_n) \leq (\limsup a_n)(\limsup b_n)$ if $a_n, b_n \geq 0$;
- $\liminf a_n \leq \limsup a_n$, with equality $\iff \lim a_n$ exists;
- for bounded sequences, the set of subsequential limits equals $[\liminf a_n, \limsup a_n]$.

1.1.3 Toeplitz transformation

Statement. $a_{nk} \geq 0$, $\sum_{k=1}^n a_{nk} = 1$, $\lim_{n \rightarrow \infty} a_{nk} = 0$ for each k . If $x_n \rightarrow L$, then $\sum_{k=1}^n a_{nk} x_k \rightarrow L$.

When to use. Weighted means and generalized Cesàro summability. Includes ordinary Cesàro ($a_{nk} = 1/n$) and many other averaging schemes.

1.2 Mean value theorems, Taylor, L'Hôpital

1.2.1 Rolle, Lagrange, Cauchy (MVT)

Statement. $f, g \in C[a, b]$, differentiable on (a, b) .

- *Rolle:* $f(a) = f(b) \Rightarrow \exists \xi \in (a, b) : f'(\xi) = 0$.
- *Lagrange:* $f(b) - f(a) = f'(\xi)(b - a)$.
- *Cauchy:* $(f(b) - f(a))g'(\xi) = (g(b) - g(a))f'(\xi)$.

When to use. Estimating increments, proving existence of points with a given derivative property, foundation for Taylor and L'Hôpital.

Auxiliary function method. To prove $\exists \xi : \Phi(\xi, f, f', \dots) = 0$, construct g with $g(a) = g(b)$ such that $g' = 0 \iff \Phi = 0$, then apply Rolle. Typical templates: $f' - \lambda f = 0 \rightarrow g = f e^{-\lambda x}$; $f' - f/x = 0 \rightarrow g = f/x$.

1.2.2 Taylor's formula

Statement. $f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k + R_n(x)$.

Remainder forms:

- *Lagrange:* $R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - a)^{n+1}$.
- *Cauchy:* $R_n = \frac{f^{(n+1)}(\xi)}{n!} (x - \xi)^n (x - a)$.
- *Integral:* $R_n = \frac{1}{n!} \int_a^x (x - t)^n f^{(n+1)}(t) dt$.
- *Peano:* $R_n = o((x - a)^n)$ as $x \rightarrow a$.

When to use. Estimates via derivatives, asymptotics, inequalities, extrema.

1.2.3 L'Hôpital's rule

Statement. f, g differentiable near a , $g'(x) \neq 0$. If $\lim f/g$ is $\frac{0}{0}$ or $\frac{\infty}{\infty}$ and $\lim f'/g' = L$, then $\lim f/g = L$.

Important. The limit of f'/g' must exist (the converse does not hold). Also works for $*/\infty$ (without requiring $f \rightarrow \infty$).

1.2.4 Darboux's theorem

Statement. f' (even if discontinuous) has the intermediate value property: if f is differentiable on $[a, b]$, then f' takes every value between $f'(a)$ and $f'(b)$.

5.4 Field extensions

5.4.1 Minimal polynomial

For algebraic $\alpha \in L/K$ there is a unique monic $m_\alpha \in K[x]$ of smallest degree with $m_\alpha(\alpha) = 0$. Properties: m_α is irreducible over K ; $K[\alpha] = K(\alpha)$, $[K(\alpha) : K] = \deg m_\alpha$; $\{1, \alpha, \dots, \alpha^{n-1}\}$ is a basis.

5.4.2 Conjugates and symmetric functions

Let $m_\alpha = \prod_{i=1}^n (x - \alpha_i)$ (separable). For any $f \in K[x]$: $\sum f(\alpha_i) \in K$, $\prod f(\alpha_i) \in K$.

5.4.3 Algebraic integers

If an algebraic integer is rational, it is an integer.

5.4.4 Trace and norm

For a separable L/K of degree n with embeddings σ_i :

$$\mathrm{Tr}_{L/K}(\alpha) = \sum \sigma_i(\alpha), \quad \mathrm{N}_{L/K}(\alpha) = \prod \sigma_i(\alpha).$$

For $L = K(\alpha)$ with $m_\alpha = x^n + c_{n-1}x^{n-1} + \dots + c_0$: $\mathrm{Tr}(\alpha) = -c_{n-1}$, $\mathrm{N}(\alpha) = (-1)^n c_0$. Transitivity: $\mathrm{Tr}_{M/K} = \mathrm{Tr}_{L/K} \circ \mathrm{Tr}_{M/L}$.

5.4.5 Structure of $\mathbb{F}_q$

For $q = p^n$ there exists a unique field $\mathbb{F}_q$. $\mathbb{F}_q = \{a \in \overline{\mathbb{F}_p} : a^q = a\}$ is the splitting field of $x^q - x$. $\mathbb{F}_q^\times$ is cyclic.

5.4.6 Cyclotomic polynomial Φ_n

$$\Phi_n(x) = \prod_{\substack{1 \leq k \leq n \\ \gcd(k, n) = 1}} (x - \zeta_n^k) \in \mathbb{Z}[x], \quad \deg \Phi_n = \varphi(n).$$

2.5.2 Residue theorem

$f \in H(U \setminus S)$, γ null-homotopic in U , avoids S , finite $S \cap \mathrm{int} \gamma$:

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{a \in S} \mathrm{Res}(f, a) \mathrm{ind}_{\gamma}(a).$$

When to use. Computing real integrals $\int_{-\infty}^{\infty} R(x) dx$, $\int_0^{2\pi} R(\cos \theta, \sin \theta) d\theta$, sums $\sum_{n \in \mathbb{Z}} f(n)$ via $\pi \cot \pi z$.

2.6 Maximum modulus and Schwarz lemma

2.6.1 Maximum modulus principle

$f \in H(U)$, $|f|$ attains a local maximum at $a \in U \Rightarrow f$ is constant on the component. Minimum principle: if $f \neq 0$, $|f|$ attains a minimum $\Rightarrow f$ constant.

Corollary (Hadamard's three circles): $\ln M(r)$ is a convex function of $\ln r$, where $M(r) = \max_{|z|=r} |f(z)|$.

2.6.2 Schwarz lemma

$f : \Delta_1(0) \rightarrow \Delta_1(0)$, $f(0) = 0 \Rightarrow |f(z)| \leq |z|$ and $|f'(0)| \leq 1$. Equality at $z \neq 0$ or $|f'(0)| = 1 \Rightarrow f(z) = e^{i\theta} z$.

5.3.4 “Too many roots” principle

$f \in \mathbb{F}_q[x]$ of degree n has at most n roots.

5.3.5 Cyclicity of $\mathbb{F}_q^\times$

$\mathbb{F}_q^\times$ is cyclic of order $q - 1$.

5.3.6 Universal factorization of $x^{q^n} - x$

$$x^{q^n} - x = \prod_{d|n} \prod_{\substack{f \text{ monic irreducible} \\ \deg f = d}} f(x).$$

The number of monic irreducibles of degree n over $\mathbb{F}_q$:

$$N_q(n) = \frac{1}{n} \sum_{d|n} \mu(d) q^{n/d} > 0.$$

5.3.7 Chevalley–Warning

$f_1, \dots, f_r \in \mathbb{F}_q[x_1, \dots, x_n]$, $\sum \deg f_i < n$. The number of common zeros in $\mathbb{F}_q^n$ is divisible by p . Consequence: if all f_i are homogeneous with no constant term, there is a non-trivial common zero (see also [3.12.3](#)).

5.3.8 Combinatorial Nullstellensatz (Alon)

F is a field, $S_i \subset F$ finite, $f \in F[x_1, \dots, x_n]$, $\deg f \leq \sum t_i$ with $t_i < |S_i|$. If the coefficient of f at $\prod x_i^{t_i}$ is non-zero, then f takes a non-zero value on $S_1 \times \dots \times S_n$ (see also [3.12.4](#)).

5.3.9 Erdős–Ginzburg–Ziv

Among any $2n - 1$ integers there are n whose sum is divisible by n (see also [7.5.5](#)).

5.2.6 Descartes' rule of signs

The number of positive roots of $f \in \mathbb{R}[x]$ (counted with multiplicity) equals the number of sign changes in the coefficient sequence $V(f)$ or is less by an even number.

5.2.7 Root bounds

1. *Cauchy*: all roots of f satisfy $|z| \leq 1 + \max_{k < n} |a_k/a_n|$.
2. *Lagrange*: all roots satisfy $|z| \leq \max(1, \sum_{k=0}^{n-1} |a_k/a_n|)$.
3. *Eneström–Kakeya*: if $0 < a_0 \leq a_1 \leq \dots \leq a_n$, then all roots lie in $|z| \leq 1$.

5.3 Polynomials over $\mathbb{F}_p$ and $\mathbb{F}_q$

5.3.1 Frobenius endomorphism

In characteristic p : $(x + y)^p = x^p + y^p$. The map $\varphi : a \mapsto a^p$ is a ring homomorphism. Over $\mathbb{F}_q$ ($q = p^n$) it is an automorphism generating $\text{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$.

5.3.2 Identity $x^q - x = \prod_{a \in \mathbb{F}_q} (x - a)$

$x^q - x = \prod_{a \in \mathbb{F}_q} (x - a)$, $x^{q-1} - 1 = \prod_{a \in \mathbb{F}_q^\times} (x - a)$. Consequence: $\gcd(f, x^q - x) = \prod_{a: f(a)=0} (x - a)$.

5.3.3 Power sums $\sum_{a \in \mathbb{F}_q} a^k$

$$\sum_{a \in \mathbb{F}_q} a^k = \begin{cases} -1, & (q-1) \mid k, \\ 0, & \text{otherwise.} \end{cases}$$

Chapter 3

Linear Algebra

Notation. $M_{m \times n}(F)$ — $m \times n$ matrices over field F ; $M_n(F)$ for square. A^T — transpose; A^* — conjugate transpose. $\text{tr}(A)$ — trace; $\det A$ — determinant; $\text{rank}(A)$ — rank. $\chi_A(t) = \det(tI - A)$ — characteristic polynomial; μ_A — minimal polynomial. $\text{Spec}(A) = \sigma(A)$ — spectrum (set of eigenvalues). $A \succ 0$ / $A \succeq 0$ — positive definite / positive semidefinite. U is *unitary* if $U^*U = I$; *orthogonal* if $Q^T Q = I$. I_n (or I) — identity matrix; $\text{diag}(\lambda_1, \dots, \lambda_n)$ — diagonal matrix. $A \otimes B$ — Kronecker product; $\text{vec}(X)$ — vectorization. $\|A\|_F$ — Frobenius norm; $\rho(A)$ — spectral radius. $\mathbb{F}_q$ — finite field of order $q = p^n$.

3.1 Characteristic and minimal polynomial, Cayley–Hamilton

3.1.1 Cayley–Hamilton theorem

Statement. For any square matrix A over a commutative ring: $\chi_A(A) = 0$, where $\chi_A(t) = \det(tI - A)$.

When to use. Reducing A^k ($k \geq n$) to linear combinations of $I, A, \dots, A^{n-1}$. Expressing A^{-1} as a polynomial in A .

3.1.2 Minimal polynomial μ_A

Statement. The monic polynomial of smallest degree with $\mu_A(A) = 0$. It divides χ_A , has the same roots. The multiplicity of λ in μ_A equals the size of the largest Jordan block for λ .

Diagonalizability criterion. A is diagonalizable over $F \iff \mu_A$ factors over F into distinct linear factors.

3.2 Trace and spectrum of products

3.2.1 Trace properties

$\text{tr}(AB) = \text{tr}(BA)$, $\text{tr}(A) = \sum \lambda_i$, $\text{tr}(A^k) = \sum \lambda_i^k$. The trace is invariant under similarity.

3.2.2 Polynomial of a matrix

If λ is an eigenvalue of A , then $p(\lambda)$ is an eigenvalue of $p(A)$ for any polynomial p . Conversely, every eigenvalue of $p(A)$ is $p(\lambda)$ for some eigenvalue λ of A . If A is diagonalizable, so is $p(A)$. Consequence: $\det p(A) = \prod p(\lambda_i)$, $\text{tr } p(A) = \sum p(\lambda_i)$.

3.2.3 Spectrum of AB vs BA

$A \in M_{m \times n}$, $B \in M_{n \times m}$: $\det(\lambda I_n - BA) = \lambda^{n-m} \det(\lambda I_m - AB)$. Non-zero eigenvalues of AB and BA coincide. When $m = n$: $\chi_{AB} = \chi_{BA}$. Corollary: $\det(I + AB) = \det(I + BA)$.

3.2.4 Newton's identities

$p_k = \text{tr}(A^k)$, $\chi_A(t) = \sum (-1)^k e_k t^{n-k}$. Then

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \cdots + (-1)^{k-1} k e_k = 0, \quad k = 1, \dots, n.$$

Over a field of characteristic 0, knowing $p_1, \dots, p_n$ recovers χ_A .

5.2 Vieta's formulas, symmetric polynomials, Newton's identities

5.2.1 Vieta's formulas

$f(x) = a_n x^n + \cdots + a_0$ with roots $\alpha_1, \dots, \alpha_n$:

$$e_k(\alpha_1, \dots, \alpha_n) = (-1)^k \frac{a_{n-k}}{a_n}.$$

In particular, $\sum \alpha_i = -a_{n-1}/a_n$, $\prod \alpha_i = (-1)^n a_0/a_n$.

5.2.2 Newton–Girard identities

$p_k = \sum \alpha_i^k$. For $1 \leq k \leq n$:

$$p_k = e_1 p_{k-1} - e_2 p_{k-2} + \cdots + (-1)^{k-1} k e_k.$$

For $k > n$: $p_k = e_1 p_{k-1} - \cdots + (-1)^{n-1} e_n p_{k-n}$.

5.2.3 Fundamental theorem of symmetric polynomials

Every symmetric $\Phi \in R[x_1, \dots, x_n]$ can be expressed uniquely in terms of $e_1, \dots, e_n$.

5.2.4 Logarithmic derivative

$\frac{f'(x)}{f(x)} = \sum_i \frac{1}{x - \alpha_i}$. Consequence: $(n-1)(f')^2 \geq n f f''$ when all roots are real.

5.2.5 Newton and Maclaurin inequalities

If all roots of $f(x) = \prod (x + \alpha_i)$ are real, then $E_k = e_k / \binom{n}{k}$ satisfy $E_k^2 \geq E_{k-1} E_{k+1}$ (log-concavity). For $\alpha_i > 0$: $S_1 \geq S_2^{1/2} \geq \cdots \geq S_n^{1/n}$, where $S_k = e_k / \binom{n}{k}$.

Idea. The Sylvester matrix is $(m+n) \times (m+n)$: first m rows contain shifted coefficients of f , next n rows contain shifted coefficients of g :

5.1.2 Resultant via roots

$f = a_n \prod (x - \alpha_i)$, $g = b_m \prod (x - \beta_j)$:

$$\text{Res}(f, g) = a_n^m b_m^n \prod_{i,j} (\alpha_i - \beta_j) = a_n^m \prod_i g(\alpha_i) = (-1)^{nm} b_m^n \prod_j f(\beta_j).$$

Properties. Multiplicativity; $\text{Res}(g, f) = (-1)^{nm} \text{Res}(f, g)$; $\text{Res}(f, x-a) = f(a)$.

5.1.3 Discriminant

$$\Delta(f) = \frac{(-1)^{n(n-1)/2}}{a_n} \text{Res}(f, f') = a_n^{2n-2} \prod_{i < j} (\alpha_i - \alpha_j)^2.$$

$$\Delta(f) = 0 \iff f \text{ has a multiple root. } \Delta(f(x+c)) = \Delta(f),$$

$$\Delta(cf) = c^{2n-2} \Delta(f).$$

5.1.4 Discriminant of $x^n + ax + b$

$$\Delta(x^n + ax + b) = (-1)^{n(n-1)/2} ((1-n)^{n-1} a^n + n^n b^{n-1}).$$

Check: $n = 2$: $a^2 - 4b$; $n = 3$: $-(4a^3 + 27b^2)$.

5.1.5 Sign of the discriminant and number of real roots

$f \in \mathbb{R}[x]$ of degree n with no multiple roots. Let s be the number of pairs of complex conjugate roots. Then $\text{sign } \Delta(f) = (-1)^s$.

3.2.5 Nilpotence criterion

Over a field of characteristic 0: A is nilpotent $\iff \text{tr}(A^k) = 0$ for $k = 1, \dots, n$.

3.3 Rank

3.3.1 Sylvester and Frobenius rank inequalities

Sylvester: $\text{rank}(A) + \text{rank}(B) - n \leq \text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.
Frobenius: $\text{rank}(AB) + \text{rank}(BC) \leq \text{rank}(B) + \text{rank}(ABC)$. Corollary: $AB = 0 \Rightarrow \text{rank}(A) + \text{rank}(B) \leq n$.

3.3.2 Rank and zero eigenvalues

For a normal (in particular Hermitian) matrix: $\text{rank}(A) = n - \dim \ker A$ equals the number of non-zero eigenvalues (counted with multiplicity). For a general square matrix this fails (a Jordan block $J_n(0)$ has rank $n-1$ but all eigenvalues are zero). In general $\text{rank}(A) \geq$ the number of non-zero eigenvalues, with equality when A is diagonalizable. For singular values: $\text{rank}(A) = \#\{\sigma_i > 0\}$ always holds.

3.3.3 Basic eigenvalue facts

- *Sum = trace:* $\sum \lambda_i = \text{tr}(A)$. *Product = determinant:* $\prod \lambda_i = \det A$.
- *Rank 1:* $A = uv^T$ has eigenvalues $\{v^T u, 0, \dots, 0\}$.
- *Nilpotent:* $A^n = 0 \iff$ all eigenvalues are 0 (over $\mathbb{C}$). More generally, A is nilpotent $\iff \chi_A(t) = t^n$.
- *Idempotent (projector):* $P^2 = P \Rightarrow$ eigenvalues $\in \{0, 1\}$.
- *Involution:* $A^2 = I$ ($\text{char} \neq 2$) $\Rightarrow$ eigenvalues $\in \{\pm 1\}$.

- *Unitary:* $U^*U = I \Rightarrow |\lambda_i| = 1$.
- *Hermitian:* $A = A^* \Rightarrow \lambda_i \in \mathbb{R}$.
- *Skew-Hermitian:* $A = -A^* \Rightarrow \lambda_i$ are purely imaginary.
- *Triangular:* eigenvalues = diagonal entries.
- *Similarity:* $S^{-1}AS$ has the same eigenvalues as A .
- *Transpose:* A and A^T have the same eigenvalues.
- *Polynomial:* $p(A)$ has eigenvalues $p(\lambda_i)$ (see 3.2.2).

3.4 Jordan form and spectral theorem

3.4.1 Jordan normal form

Over $\mathbb{C}$: $A = SJS^{-1}$, $J = \bigoplus J_{m_k}(\lambda_k)$, $J_m(\lambda) = \lambda I_m + N_m$. The number of blocks of size $\geq k$ for λ equals $\dim \ker(A - \lambda I)^k - \dim \ker(A - \lambda I)^{k-1}$.

3.4.2 Spectral theorem for normal matrices

$A \in M_n(\mathbb{C})$ is normal ($AA^* = A^*A$) $\iff$ there exists a unitary U with $U^*AU = \text{diag}(\lambda_1, \dots, \lambda_n)$.

- Hermitian ($A = A^*$): $\lambda_i \in \mathbb{R}$.
- Unitary ($A^*A = I$): $|\lambda_i| = 1$.
- Real symmetric: orthogonally diagonalizable over $\mathbb{R}$.

3.4.3 Schur triangularization

Over $\mathbb{C}$: there exists a unitary U and upper-triangular T with $U^*AU = T$, $T_{ii} = \lambda_i$.

Chapter 5

Polynomials, Rings and Fields

Notation. $K[x]$ — polynomials over field K ; $\overline{K}$ — algebraic closure. $\deg f$ — degree; f' — formal derivative ($f' = \sum ia_i x^{i-1}$, with the usual rules). $\text{Res}(f, g)$ — resultant; $\Delta(f)$ — discriminant. e_k — elementary symmetric sums; $p_k = \sum \alpha_i^k$ — power sums. $\mathbb{F}_q$ — finite field of order $q = p^n$. $[L : K]$ — degree of field extension. $\Phi_n(x)$ — n -th cyclotomic polynomial. A polynomial $f \in L[x]$ is *separable* if it has no repeated roots; a field extension L/K is separable if every algebraic element has a separable minimal polynomial.

5.1 Resultant and discriminant

5.1.1 Resultant — definition and common root criterion

Statement. For $f(x) = a_n x^n + \dots + a_0$, $g(x) = b_m x^m + \dots + b_0$ over an integral domain: $\text{Res}(f, g) = \det \text{Syl}(f, g)$.

$\text{Res}(f, g) = 0 \iff f, g$ have a common divisor of degree ≥ 1 in $\overline{K}[x]$.

4.4.8 Index and k -th power criterion

p odd, g a primitive root mod p . The map $\text{ind}_g : (\mathbb{Z}/p)^\times \rightarrow \mathbb{Z}/(p-1)$ is an isomorphism. a is a k -th power $\iff a^{(p-1)/\gcd(k, p-1)} \equiv 1 \pmod{p}$. The number of k -th powers in $(\mathbb{Z}/p)^\times$ is $(p-1)/\gcd(k, p-1)$.

4.4.9 Power sums modulo a prime

For a prime p and $m \geq 0$:

$$\sum_{a=1}^{p-1} a^m \equiv \begin{cases} -1 \pmod{p}, & m \geq 1 \text{ and } (p-1) \mid m, \\ 0 \pmod{p}, & \text{otherwise.} \end{cases}$$

4.4.10 Generalized Wilson theorem

$$\prod_{\substack{1 \leq a \leq n \\ \gcd(a, n)=1}} a \equiv \begin{cases} -1 \pmod{n}, & n \in \{2, 4, p^k, 2p^k\}, \\ +1 \pmod{n}, & \text{otherwise.} \end{cases}$$

3.5 Positive (semi)definite matrices

3.5.1 Equivalent criteria for PSD

For a Hermitian A the following are equivalent:

1. $x^*Ax \geq 0$ for all x ;
2. $\lambda_i \geq 0$;
3. $A = B^*B$ for some B ;
4. all principal minors ≥ 0 .

For PD: strict inequalities; in (4) it suffices to check leading principal minors (Sylvester's criterion).

3.5.2 Gram matrix representation

$A \succeq 0 \iff A = B^*B \iff A_{ij} = \langle v_i, v_j \rangle$. $\text{rank}(A) = \dim \text{span}(v_i)$. Every PSD matrix is the Gram matrix of some set of vectors.

3.5.3 Cholesky decomposition

$A \succ 0 \iff$ there exists a unique lower-triangular L with $L_{ii} > 0$ and $A = LL^*$.

3.5.4 Square root

For $A \succeq 0$ there exists a unique $A^{1/2} \succeq 0$ with $(A^{1/2})^2 = A$ (via the spectrum: $A^{1/2} = U\Lambda^{1/2}U^*$).

3.5.5 Schur complement

For $M = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}$ with $D \succ 0$: $M \succ 0 \iff A - BD^{-1}B^* \succ 0$.

3.5.6 Löwner order

$A \succeq B \iff A - B \succeq 0$. Properties: $A \succeq B \Rightarrow C^*AC \succeq C^*BC$; $0 \prec A \preceq B \Rightarrow B^{-1} \preceq A^{-1}$.

3.5.7 Singular value decomposition (SVD)

Any $A \in \mathbb{C}^{m \times n}$: $A = U\Sigma V^*$, U, V unitary, Σ diagonal with $\sigma_1 \geq \dots \geq \sigma_r > 0$. $\sigma_i = \sqrt{\lambda_i(A^*A)}$, $r = \text{rank}(A)$.

3.5.8 Polar decomposition

Any A : $A = UP$, U unitary, $P = (A^*A)^{1/2} \succeq 0$.

3.6 Euclidean spaces and Gram matrices

3.6.1 Gram matrix

$G_{ij} = \langle v_i, v_j \rangle$. G is symmetric and PSD. $G \succ 0 \iff$ the v_i are linearly independent. $\det G = (\text{Vol})^2$.

3.6.2 Hadamard's inequality

$|\det A| \leq \prod \|a_i\|$ (column vectors). Equality $\iff$ columns are orthogonal. For PSD: $\det H \leq \prod H_{ii}$.

3.6.3 Orthogonal projection

For $W \subseteq V$: $P_W^2 = P_W$, $P_W^T = P_W$, $\text{im } P_W = W$, $\ker P_W = W^\perp$. $\|x - P_W x\| = \min_{y \in W} \|x - y\|$. If the columns of A form a basis of W : $P_W = A(A^T A)^{-1} A^T$.

4.4.3 Structure of $(\mathbb{Z}/n)^\times$

$(\mathbb{Z}/n)^\times$ is cyclic $\iff n \in \{1, 2, 4, p^k, 2p^k\}$ (p odd prime). For odd p : $(\mathbb{Z}/p^k)^\times \cong \mathbb{Z}/p^{k-1}(p-1)$. For $p = 2$, $k \geq 3$: $(\mathbb{Z}/2^k)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{k-2}$.

4.4.4 Smallest prime divisor trick

$n \geq 2$, p the smallest prime divisor of n , $\gcd(a, n) = 1$.

- If $n \mid a^n - 1$, then $a \equiv 1 \pmod{p}$.
- If n is odd and $n \mid a^n + 1$, then $a \equiv -1 \pmod{p}$.

When to use. IMO 1990-3 ($n^2 \mid 2^n + 1$), problem “there is no $n > 1$ with $n \mid 2^n - 1$ ”.

4.4.5 Carmichael function

$$\lambda(p^k) = \begin{cases} p^{k-1}(p-1), & p \text{ odd or } p^k \in \{2, 4\}, \\ 2^{k-2}, & p = 2, k \geq 3, \end{cases}$$

$$\lambda(n) = \text{lcm}(\lambda(p_1^{a_1}), \dots, \lambda(p_r^{a_r})).$$

In $(\mathbb{Z}/n)^\times$ there always exists an element of order $\lambda(n)$.

4.4.6 Prime divisors of cyclotomic polynomials

$p \nmid a$. If $p \mid \Phi_n(a)$, then either $\text{ord}_p(a) = n$ ($\implies p \equiv 1 \pmod{n}$), or $p \mid n$.

4.4.7 Infinitely many primes $\equiv 1 \pmod{n}$

For any $n \geq 1$ there are infinitely many primes $p \equiv 1 \pmod{n}$. Idea: take a prime divisor of $\Phi_n(M)$ (see 4.4.6).

4.3.4 Structure of squares in $(\mathbb{Z}/p^k)^\times$

p odd, $\gcd(a, p) = 1$: a is a square in $(\mathbb{Z}/p^k)^\times \iff \left(\frac{a}{p}\right) = +1$.
For $p = 2$ ($k \geq 3$): odd a is a square modulo $2^k \iff a \equiv 1 \pmod{8}$.

4.3.5 Sum of Legendre symbols

With $p \nmid a$: $\sum_{x=0}^{p-1} \left(\frac{ax+b}{p}\right) = 0$. For $f(x) = ax^2 + bx + c$, $p \nmid a$, $D = b^2 - 4ac$:

$$\sum_{x=0}^{p-1} \left(\frac{f(x)}{p}\right) = \begin{cases} -\left(\frac{a}{p}\right), & p \nmid D, \\ (p-1)\left(\frac{a}{p}\right), & p \mid D. \end{cases}$$

4.4 Order of an element, primitive roots

4.4.1 Properties of order

$\gcd(a, n) = 1$, $d = \text{ord}_n(a)$.

1. $a^k \equiv 1 \pmod{n} \iff d \mid k$.
2. $d \mid \varphi(n)$.
3. $\text{ord}_n(a^k) = d / \gcd(d, k)$.
4. If $\gcd(\text{ord}_n a, \text{ord}_n b) = 1$, then $\text{ord}_n(ab) = \text{ord}_n(a) \cdot \text{ord}_n(b)$.
5. In an abelian group there exists an element of order $\text{lcm}(\text{ord}_n a, \text{ord}_n b)$.

4.4.2 Existence of primitive roots

For any prime p , the group $(\mathbb{Z}/p)^\times$ is cyclic. The number of primitive roots mod p is $\varphi(p-1)$. *Test.* g is a primitive root $\iff g^{(p-1)/q} \not\equiv 1 \pmod{p}$ for every prime $q \mid p-1$.

3.6.4 Bessel's inequality

For an orthonormal system $e_1, \dots, e_k$: $\sum_{i=1}^k |\langle x, e_i \rangle|^2 \leq \|x\|^2$. Equality $\iff x \in \text{span}(e_i)$.

3.6.5 Cauchy–Schwarz inequality

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$, equality $\iff u, v$ are linearly dependent (see also 1.3.2).

3.7 Analytic geometry

3.7.1 Line in the plane

- *Parametric*: $(x, y) = (x_0, y_0) + t(a, b)$.
- *General equation*: $Ax + By + C = 0$, normal (A, B) .
- *Distance from point* (x_0, y_0) : $\frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$.
- *Angle between lines*: $\cos \theta = \frac{|A_1 A_2 + B_1 B_2|}{\sqrt{A_1^2 + B_1^2} \sqrt{A_2^2 + B_2^2}}$.

3.7.2 Plane and line in space

- *Plane*: $Ax + By + Cz + D = 0$, normal (A, B, C) .
- *Distance from point to plane*: $\frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$.
- *Line in $\mathbb{R}^3$* : $(x, y, z) = (x_0, y_0, z_0) + t(a, b, c)$.
- *Distance between skew lines*: $d = \frac{|(\mathbf{r}_2 - \mathbf{r}_1) \cdot (\mathbf{a}_1 \times \mathbf{a}_2)|}{\|\mathbf{a}_1 \times \mathbf{a}_2\|}$.

3.7.3 Conic sections

General second-degree equation: $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

The type is determined by the discriminant $\Delta = B^2 - 4AC$:

- $\Delta < 0$: ellipse (circle when $A = C$, $B = 0$).
- $\Delta = 0$: parabola.
- $\Delta > 0$: hyperbola.

Ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a \geq b > 0.$$

- Foci $(\pm c, 0)$, $c = \sqrt{a^2 - b^2}$.
- $r_1 + r_2 = 2a$ (sum of distances to foci is constant).
- Eccentricity $e = c/a < 1$.
- Directrices $x = \pm a/e$.
- Area $S = \pi ab$.

Hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

- Foci $(\pm c, 0)$, $c = \sqrt{a^2 + b^2}$.
- $|r_1 - r_2| = 2a$ (absolute difference of distances to foci is constant).
- Eccentricity $e = c/a > 1$.
- Directrices $x = \pm a/e$.
- Asymptotes $y = \pm(b/a)x$.

4.2.9 Pell equations

$D > 0$ is not a square. All positive solutions of $x^2 - Dy^2 = 1$:

$$x_n + y_n\sqrt{D} = (x_1 + y_1\sqrt{D})^n,$$

where (x_1, y_1) is the minimal positive solution.

4.3 Quadratic residues

4.3.1 Euler's criterion

For odd prime p and $p \nmid a$:

$$a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}.$$

Corollary: $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$.

4.3.2 Quadratic reciprocity law

For distinct odd primes p, q :

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Supplementary.

$$\left(\frac{-1}{p}\right) = \begin{cases} +1, & p \equiv 1 \pmod{4}, \\ -1, & p \equiv 3 \pmod{4}. \end{cases}$$

$$\left(\frac{2}{p}\right) = \begin{cases} +1, & p \equiv \pm 1 \pmod{8}, \\ -1, & p \equiv \pm 3 \pmod{8}. \end{cases}$$

4.3.3 Jacobi symbol

For odd $n = \prod p_i^{a_i}$: $\left(\frac{a}{n}\right) = \prod \left(\frac{a}{p_i}\right)^{a_i}$. It is multiplicative in both arguments; the same reciprocity law holds. Important: $\left(\frac{a}{n}\right) = 1$ does not imply a is a square modulo n .

4.2.3 p -adic balance

If $A = B$ in $\mathbb{Z}$, then $v_p(A) = v_p(B)$ for every p . Usually combined with Legendre's formula (4.1.4) and LTE (4.1.1).

4.2.4 Hensel's lemma

Let $f \in \mathbb{Z}[x]$, p prime. If $f(a) \equiv 0 \pmod{p}$ and $f'(a) \not\equiv 0 \pmod{p}$, then for each $k \geq 1$ there exists a unique $a_k \in \mathbb{Z}/p^k$ with $a_k \equiv a \pmod{p}$ and $f(a_k) \equiv 0 \pmod{p^k}$.

4.2.5 Chinese Remainder Theorem (CRT)

$n_1, \dots, n_k$ are pairwise coprime. For any a_i there exists x with $x \equiv a_i \pmod{n_i}$, unique modulo $\prod n_i$.

4.2.6 Vieta jumping

The equation is symmetric in a, b and quadratic in a for fixed b . If (a, b) is a solution, then the second root a' is also an integer solution. Usually $|a'| < |a|$, giving a descent.

4.2.7 Sophie Germain identity

$$a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2).$$

When to use. Factorization of $n^4 + 4$ and $n^4 + 4^n$ (for odd n).

4.2.8 Order obstruction

If $\gcd(a, p) = 1$ and $d = \text{ord}_p(a)$, then $a^n \equiv 1 \pmod{p} \iff d \mid n$. By Fermat's little theorem $d \mid p - 1$.

Parabola

$$y^2 = 2px, p > 0.$$

- Focus $(p/2, 0)$.
- Directrix $x = -p/2$.
- $r = d$ (distance to focus equals distance to directrix).
- Eccentricity $e = 1$.

3.8 Bilinear and quadratic forms. Signature

3.8.1 Sylvester's law of inertia

For a symmetric $A \in \mathbb{R}^{n \times n}$ the signature (p, q, r) — the numbers of positive, negative, and zero eigenvalues — is invariant under congruence $A \mapsto C^T A C$.

3.8.2 Simultaneous diagonalization

If $A \succ 0$ and B is symmetric, then there exists C with $C^T A C = I$, $C^T B C = \text{diag}(\mu_i)$. The numbers μ_i are the roots of $\det(B - \mu A) = 0$, all real.

3.9 Determinants

3.9.1 Vandermonde

$\det(x_j^{i-1}) = \prod_{i < j} (x_j - x_i)$. Generalization: for monic polynomials P_{i-1} of degree $i - 1$ the determinant is the same.

3.9.2 Block matrix formulas

$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \cdot \det(D - CA^{-1}B)$ (when A is invertible). If $CD = DC$, then $\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(AD - BC)$.

3.9.3 Cauchy–Binet

For $A \in M_{m \times n}$, $B \in M_{n \times m}$, $m \leq n$:

$$\det(AB) = \sum_{S, |S|=m} \det(A_{[m],S}) \cdot \det(B_{S,[m]}).$$

Corollary: $\det(A^T A) = \sum_S \det(A_S)^2 \geq 0$.

3.9.4 Sylvester’s identity

$\det(I_m + AB) = \det(I_n + BA)$. Special case: $\det(I + uv^T) = 1 + v^T u$.

3.9.5 Rank-one perturbation

$\det(A + uv^T) = \det(A) \cdot (1 + v^T A^{-1}u)$. For rank k : $\det(A + UV^T) = \det(A) \cdot \det(I_k + V^T A^{-1}U)$.

3.9.6 Cauchy determinant

$$\det(1/(a_i + b_j)) = \frac{\prod_{i < j} (a_i - a_j)(b_i - b_j)}{\prod_{i,j} (a_i + b_j)}.$$

3.9.7 Circulant and the DFT matrix

$C = \text{circ}(c_0, \dots, c_{n-1})$ is circulant: each row is the previous row shifted right. With $\omega = e^{2\pi i/n}$, C is diagonalized by the DFT matrix:

$$F_{jk} = \frac{1}{\sqrt{n}} \omega^{jk}, \quad j, k = 0, \dots, n-1.$$

4.1.7 Zsigmondy’s theorem

$a > b \geq 1$, $\gcd(a, b) = 1$, $n \geq 2$. The number $a^n - b^n$ has a prime divisor that does not divide $a^k - b^k$ for any $k < n$ (primitive). Exceptions:

- $n = 2$ and $a + b$ is a power of 2;
- $(a, b, n) = (2, 1, 6)$.

Analog for sum: for $n \geq 2$, $a > b \geq 1$, $\gcd(a, b) = 1$, $a^n + b^n$ has a primitive prime divisor, except for $n = 2$, $a + b$ a power of 2, or $(a, b, n) = (2, 1, 3)$.

4.2 Diophantine tricks: descent, moduli, estimates

4.2.1 Infinite descent

Idea. If from any integer solution one can construct a solution with strictly smaller positive measure, then no solutions exist.

When to use. Equations like $x^4 + y^4 = z^2$, $x^2 + y^2 = 3z^2$, “there is no $n > 1$ with $n \mid 2^n - 1$ ”.

4.2.2 Modulo obstruction

If the equation $F = 0$ has no solutions modulo n , then it has no solutions in $\mathbb{Z}$ either.

Useful tables.

$$x^2 \bmod 3 \in \{0, 1\}, \quad x^2 \bmod 4 \in \{0, 1\}, \quad x^2 \bmod 8 \in \{0, 1, 4\}.$$

Consequences: $a^2 + b^2 \not\equiv 3 \pmod{4}$, sum of three squares is not $\equiv 7 \pmod{8}$.

4.1.3 LTE for $p = 2$

Statement. x, y odd, $n \geq 1$.

- n odd: $v_2(x^n - y^n) = v_2(x - y)$.
- n even: $v_2(x^n - y^n) = v_2(x - y) + v_2(x + y) + v_2(n) - 1$.
- If $4 \mid x - y$, then $v_2(x^n - y^n) = v_2(x - y) + v_2(n)$.

For sum: when n is odd $v_2(x^n + y^n) = v_2(x + y)$; when n is even $v_2(x^n + y^n) = 1$.

4.1.4 Legendre's formula

$$v_p(n!) = \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor = \frac{n - s_p(n)}{p - 1},$$

where $s_p(n)$ is the sum of digits of n in base p .

4.1.5 Kummer's theorem

$v_p\binom{m+n}{n}$ equals the number of carries when adding m and n in base p . Equivalently:

$$v_p\binom{m+n}{n} = \frac{s_p(m) + s_p(n) - s_p(m+n)}{p-1}.$$

4.1.6 Lifting the order modulo p^k

Let p be odd, $p \nmid a$, $d = \text{ord}_p(a)$, $c = v_p(a^d - 1) \geq 1$. For all $k \geq 1$:

$$\text{ord}_{p^k}(a) = d \cdot p^{\max(0, k-c)}.$$

For $p = 2$: when $a \equiv 1 \pmod{4}$ we have $\text{ord}_{2^k}(a) = 2^{\max(0, k-v_2(a-1))}$.

Then $F^*CF = \text{diag}(p(\omega^0), \dots, p(\omega^{n-1}))$, $p(x) = \sum c_k x^k$. Hence $\det C = \prod p(\omega^k)$, eigenvectors $v_k = (1, \omega^k, \dots, \omega^{k(n-1)})^T$.

DFT matrix properties. F is unitary: $F^{-1} = F^*$ with $(F^{-1})_{jk} = \frac{1}{\sqrt{n}} \omega^{-jk} = \overline{F_{jk}}$. Also $F^4 = I$ (the DFT is a fourth root of the identity), so $\det(F)^4 = 1$ and thus $\det F \in \{\pm 1, \pm i\}$. The unnormalized DFT matrix $\tilde{F}_{jk} = \omega^{jk}$ (no $1/\sqrt{n}$ factor) satisfies $\tilde{F}^{-1} = \frac{1}{n} \tilde{F}^*$ and $|\det \tilde{F}| = n^{n/2}$.

F is a scaled Vandermonde matrix of roots of unity. Convolution theorem: $C \cdot v$ corresponds to convolution of (c_k) with v ; F turns convolution into pointwise multiplication.

Root-of-unity sums. $\sum_{k=0}^{n-1} \omega^{km} = n \cdot [n \mid m]$. Gauss sum: $\sum_{k=0}^{n-1} \omega^{k^2}$ equals $\sqrt{n}$ or $i\sqrt{n}$ depending on $n \pmod{4}$.

3.10 Special classes of matrices

3.10.1 Nilpotent matrix

$N^k = 0$. Equivalent: $\sigma(N) = \{0\}$, $\chi_N(t) = t^n$. Jordan form consists of blocks $J_{m_i}(0)$. $(I - N)^{-1} = I + N + \dots + N^{n-1}$.

3.10.2 Idempotent

$P^2 = P$. Then P is diagonalizable, $V = \text{im } P \oplus \ker P$, $\sigma(P) \subseteq \{0, 1\}$. $\text{rank}(P) = \text{tr}(P)$. $I - P$ is also an idempotent.

3.10.3 Involution

$A^2 = I$, $\text{char} \neq 2$. Then $\sigma(A) \subseteq \{\pm 1\}$, A is diagonalizable. $V = V_+ \oplus V_-$, $V_{\pm} = \ker(A \mp I)$. Projectors $P_{\pm} = (I \pm A)/2$.

3.10.4 Commuting matrices

If $AB = BA$, then: there exists a common eigenvector; a family of pairwise commuting matrices is *simultaneously triangularizable*.

(there exists S such that $S^{-1}A_iS$ are all upper-triangular); if all are diagonalizable — simultaneously diagonalizable.

3.10.5 Sylvester equation

$AX - XB = C$ has a unique solution $\iff \sigma(A) \cap \sigma(B) = \emptyset$.

3.10.6 Commutator

$\text{tr}(AB - BA) = 0$. If $\text{char} = 0$, the equation $AB - BA = I$ has no solution.

3.11 Kronecker product

3.11.1 Definition

For $A \in M_{m \times n}$, $B \in M_{p \times q}$ the Kronecker product $A \otimes B$ is the $mp \times nq$ block matrix

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}.$$

3.11.2 Basic algebraic properties

- *Distributivity:* $(A+B) \otimes C = A \otimes C + B \otimes C$, $A \otimes (B+C) = A \otimes B + A \otimes C$.
- *Associativity:* $A \otimes (B \otimes C) = (A \otimes B) \otimes C$.
- *Homogeneity:* $(\alpha A) \otimes B = A \otimes (\alpha B) = \alpha(A \otimes B)$.
- *Mixed-product:* $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, whenever the ordinary products are defined.

Chapter 4

Number Theory

Notation. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ — usual. $v_p(n)$ — exponent of p in n (valuation). $a \mid b$ — a divides b ; $p^k \parallel n$ means $p^k \mid n$, $p^{k+1} \nmid n$. $\text{ord}_n(a)$ — multiplicative order of a modulo n . $\varphi(n)$ — Euler totient; $\mu(n)$ — Möbius function. $\left(\frac{a}{p}\right)$ — Legendre symbol; $\left(\frac{a}{n}\right)$ — Jacobi symbol. $\mathbb{Z}/n\mathbb{Z}$ (or $\mathbb{Z}_n$) — integers modulo n .

4.1 LTE (Lifting The Exponent)

4.1.1 LTE for odd p , difference

Statement. Let p be an odd prime, $x, y \in \mathbb{Z}$, $n \geq 1$, $p \mid x - y$, $p \nmid xy$. Then

$$v_p(x^n - y^n) = v_p(x - y) + v_p(n).$$

When to use. Problems like “prove $p^k \mid a^n - b^n$ ”, “find all n with $p^k \mid a^n \pm b^n$ ”.

4.1.2 LTE for odd p , sum

Statement. p odd, n odd, $p \mid x + y$, $p \nmid xy$. Then

$$v_p(x^n + y^n) = v_p(x + y) + v_p(n).$$

3.15.3 Weyl inequalities

For Hermitian A, B : $\lambda_{i+j-1}(A+B) \leq \lambda_i(A) + \lambda_j(B)$.

- *Transpose*: $(A \otimes B)^T = A^T \otimes B^T$, $(A \otimes B)^* = A^* \otimes B^*$.
- *Inverse*: $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$ for invertible A, B .
- *Powers*: $(A \otimes B)^k = A^k \otimes B^k$ ($k \geq 0$); $e^{A \otimes B} = e^A \otimes e^B$.

3.11.3 Trace, determinant, rank, spectrum

For square $A \in M_m$, $B \in M_n$:

- $\text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B)$.
- $\det(A \otimes B) = (\det A)^n (\det B)^m$.
- $\text{rank}(A \otimes B) = \text{rank}(A) \text{rank}(B)$.
- $\sigma(A \otimes B) = \{\lambda_i \mu_j : \lambda_i \in \sigma(A), \mu_j \in \sigma(B)\}$.

3.11.4 Vectorization and Kronecker sum

$\text{vec}(AXB) = (B^T \otimes A) \text{vec}(X)$.

Kronecker sum: $A \oplus B = A \otimes I_n + I_m \otimes B$ for $A \in M_m$, $B \in M_n$. Then $\sigma(A \oplus B) = \{\lambda_i + \mu_j\}$ and $e^{A \oplus B} = e^A \otimes e^B$. Sylvester equation $AX + XB = C$ (3.10.5) rewrites as $(I \otimes A + B^T \otimes I) \text{vec } X = \text{vec } C$.

3.12 Matrices over $\mathbb{F}_p$

3.12.1 Frobenius identity

For commuting A, B over a field of char p : $(A+B)^{p^k} = A^{p^k} + B^{p^k}$.

3.12.2 Order of $\text{GL}_n(\mathbb{F}_q)$

$|\text{GL}_n(\mathbb{F}_q)| = \prod_{k=0}^{n-1} (q^n - q^k)$. Number of k -dimensional subspaces: $\binom{n}{k}_q$.

3.12.3 Chevalley–Warning theorem

$f_1, \dots, f_r \in \mathbb{F}_q[x_1, \dots, x_n]$, $\sum \deg f_i < n$. The number of common zeros in $\mathbb{F}_q^n$ is divisible by p . If there is a trivial zero, there is a non-trivial one (see also 5.3.7).

3.12.4 Combinatorial Nullstellensatz (Alon)

$f \in F[x_1, \dots, x_n]$, $\deg f = \sum t_i$, the coefficient of $\prod x_i^{t_i}$ is non-zero. For $S_i \subseteq F$ with $|S_i| > t_i$ there exists $s \in \prod S_i$ with $f(s) \neq 0$ (see also 5.3.8).

3.13 Matrix exponential

3.13.1 Computation

$$e^{tJ_m(\lambda)} = e^{t\lambda} \sum_{k=0}^{m-1} \frac{t^k}{k!} N_m^k. \quad e^{tA} = S e^{tJ} S^{-1}.$$

$$\sigma(e^A) = e^{\sigma(A)}, \det e^A = e^{\operatorname{tr} A}.$$

$$AB = BA \Rightarrow e^{A+B} = e^A e^B.$$

3.13.2 Estimates

$$\|A^n\| \leq C \cdot n^{m-1} \rho(A)^n, \quad \|e^{tA}\| \leq C(1 + t^{m-1})e^{\alpha(A)t} \quad (t \geq 0).$$

Gelfand's formula: $\rho(A) = \lim \|A^n\|^{1/n}$.

3.14 Integer matrices (Smith normal form)

3.14.1 Smith normal form

For $A \in M_{m \times n}(\mathbb{Z})$ of rank r there exist unimodular U, V ($U, V \in \operatorname{GL}_n(\mathbb{Z})$, i.e. $\det U, \det V = \pm 1$) with $UAV = \operatorname{diag}(d_1, \dots, d_r, 0, \dots)$, $d_1 \mid d_2 \mid \dots \mid d_r$. $\prod d_i = |\det A|$ (for square A).

3.14.2 Invariant factors via minors

$\Delta_k(A) = \gcd$ of all $k \times k$ minors of A . $d_k = \Delta_k / \Delta_{k-1}$.

3.14.3 Structure of the cokernel

$\mathbb{Z}^n / A\mathbb{Z}^n \cong \bigoplus \mathbb{Z}/d_i \oplus \mathbb{Z}^{n-r}$. For non-singular A : $|\mathbb{Z}^n / A\mathbb{Z}^n| = |\det A|$.

3.14.4 Connection to graphs

For a connected graph G with n vertices and m edges, let B be its $n \times m$ incidence matrix (entries ± 1 , one per edge). The SNF of B has $r = n - 1$, $d_1 = 1$, and the remaining $d_2, \dots, d_{n-1}$ are the invariant factors of the cycle space. Moreover, the SNF of the Laplacian L yields $\prod d_i = n \cdot t(G)$ where $t(G)$ is the number of spanning trees (8.2.3). The abelian sandpile (critical) group of G is $\bigoplus_{i=2}^{n-1} \mathbb{Z}/d_i$.

3.15 Minimax characterizations and eigenvalue inequalities

3.15.1 Courant–Fischer

For Hermitian A with $\lambda_1 \geq \dots \geq \lambda_n$:

$$\lambda_k = \max_{\dim V=k} \min_{x \in V, \|x\|=1} x^* A x = \min_{\dim V=n-k+1} \max_{x \in V, \|x\|=1} x^* A x.$$

For singular values: $\sigma_k(A) = \max_{\dim V=k} \min_{x \in V, \|x\|=1} \|Ax\|$.

3.15.2 Cauchy interlacing

For a principal submatrix B of size $n - 1$: $\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \dots \geq \mu_{n-1} \geq \lambda_n$.

Φ_n is irreducible over $\mathbb{Q}$. $x^n - 1 = \prod_{d|n} \Phi_d(x)$, $\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}$.

5.4.7 Tower formula

$$[L : K] = [L : F] \cdot [F : K].$$

5.4.8 Primitive element

A finite separable L/K is of the form $L = K(\theta)$. For $K(\alpha, \beta)$ with infinite K , $\theta = \alpha + c\beta$ works for almost all c .

5.4.9 Kronecker on roots of unity

If $\alpha \neq 0$ is an algebraic integer and all its conjugates satisfy $|\alpha_i| \leq 1$, then α is a root of unity.

5.5 Factorization and irreducibility

5.5.1 Gauss's lemma

The product of primitive polynomials in $\mathbb{Z}[x]$ is primitive. f is irreducible over $\mathbb{Z} \Leftrightarrow$ irreducible over $\mathbb{Q}$ (up to a constant factor).

5.5.2 Eisenstein's criterion

$f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$, p prime. If $p \nmid a_n$, $p \mid a_i$ ($0 \leq i \leq n-1$), $p^2 \nmid a_0$, then f is irreducible over $\mathbb{Q}$. *Tricks:* shift $x \mapsto x + a$; reversal $\hat{f}(x) = x^n f(1/x)$.

5.5.3 Generalized Eisenstein

If $p \mid a_0, \dots, a_k$, $p \nmid a_{k+1}$, $p^2 \nmid a_0$, then every non-constant factor of f has degree $> k$.

5.5.4 Reduction modulo p

$f \in \mathbb{Z}[x]$, $p \nmid a_n$. If $\bar{f}$ is irreducible in $\mathbb{F}_p[x]$, then f is irreducible in $\mathbb{Z}[x]$. The converse is false ($x^4 + 1$ is a counterexample).

5.5.5 Root estimate (simple constant term)

$f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$, $|a_0|$ is prime, $|a_0| > |a_1| + \cdots + |a_n|$. Then f is irreducible.

5.5.6 Irreducibility of Φ_p via shift

$\Phi_p(x+1) = \frac{(x+1)^p - 1}{x} = x^{p-1} + \binom{p}{1}x^{p-2} + \cdots + \binom{p}{p-1}$. Apply Eisenstein with p .

5.5.7 Newton polygon

$f = \sum a_i x^i \in \mathbb{Z}_p[x]$, plot points $(i, v_p(a_i))$, take lower convex hull. If the slopes are $\lambda_1, \dots, \lambda_k$ with multiplicities $m_1, \dots, m_k$, then the valuations of the roots are $-\lambda_i$ with multiplicity m_i . Irreducibility criterion: a single segment of slope m/n in lowest terms $\Rightarrow f$ remains irreducible over any extension of degree $< n$. If $v_p(a_n) = 0$, $v_p(a_0) = 1$, and every other $v_p(a_i) \geq 1$, then f is irreducible (Eisenstein via Newton).

5.5.8 Hensel lifting

$f \in \mathbb{Z}[x]$, $f(a) \equiv 0 \pmod{p}$, $f'(a) \not\equiv 0 \pmod{p}$. Then for any $k \geq 1$ there exists a unique $\tilde{a} \pmod{p^k}$ with $\tilde{a} \equiv a \pmod{p}$ and $f(\tilde{a}) \equiv 0 \pmod{p^k}$ (see also [4.2.4](#)).

Chapter 6

Group Theory

Notation. G — group; $H \leq G$ — subgroup; $H \trianglelefteq G$ — normal subgroup. $[G : H]$ — index; $|G|$ — order. $Z(G)$ — center; $C_G(H)$ — centralizer; $N_G(H)$ — normalizer. $\text{cl}(g)$ — conjugacy class of g . $\text{Aut}(G)$ — automorphism group; $\text{Inn}(G)$ — inner automorphisms. G/H — quotient group; $G \cong H$ — isomorphic. S_n — symmetric group; A_n — alternating group. $\mathbb{Z}/n\mathbb{Z}$ (or C_n) — cyclic group of order n .

6.1 Orders, Lagrange, Cauchy, cyclic groups

6.1.1 Lagrange's theorem

$|G| < \infty$, $H \leq G \Rightarrow |H| \mid |G|$ and $|G| = [G : H] \cdot |H|$. Corollary: $\text{ord}(g) \mid |G|$, $g^{|G|} = e$.

6.1.2 Cauchy's theorem

p prime, $p \mid |G| \Rightarrow \exists$ element of order p .

6.1.3 Order of a power

$\text{ord}(g) = n < \infty \Rightarrow \text{ord}(g^k) = n / \gcd(k, n)$.

6.1.4 Subgroup structure of cyclic groups

$G = \langle g \rangle$, $|G| = n$: every subgroup is cyclic; for each $d \mid n$ there is exactly one subgroup of order d : $\langle g^{n/d} \rangle$.

6.1.5 Order of product in abelian groups

G abelian, $\text{ord}(a) = m$, $\text{ord}(b) = n$:

1. $\gcd(m, n) = 1 \Rightarrow \text{ord}(ab) = mn$.
2. In $\langle a, b \rangle \exists$ an element of order $\text{lcm}(m, n)$.

Caution: in non-abelian groups this may fail (S_3).

6.1.6 Cyclicity criterion

A finite G is cyclic $\Leftrightarrow$ for each $d \mid |G|$ the equation $x^d = e$ has at most d solutions. Equivalently: for each d at most one subgroup of order d .

6.1.7 Groups of order p^2 are abelian

$|G| = p^2 \Rightarrow G$ is abelian. Up to isomorphism: $\mathbb{Z}/p^2$ and $\mathbb{Z}/p \times \mathbb{Z}/p$.

6.1.8 Groups of order pq

$|G| = pq$, $p < q$ primes. Then:

1. $p \nmid q - 1 \Rightarrow G \cong \mathbb{Z}/(pq)$.
2. $p \mid q - 1 \Rightarrow$ two groups: cyclic and $\mathbb{Z}/q \rtimes \mathbb{Z}/p$ (non-abelian semidirect product $G = N \rtimes H$ where $N \trianglelefteq G$, $NH = G$, $N \cap H = \{e\}$).

6.1.9 Subgroup of smallest prime index is normal

$[G : H] = p$ is the smallest prime divisor of $|G| \Rightarrow H \trianglelefteq G$.

8.8 Extremal graph theory

8.8.1 Turán's theorem

The maximum number of edges in an n -vertex graph with no $(r + 1)$ -clique is $(1 - \frac{1}{r}) \frac{n^2}{2}$, attained by the complete r -partite Turán graph $T_r(n)$ with parts as equal as possible.

Consequence: if $m > n^2/4$ then G contains a triangle (case $r = 2$, Mantel's theorem).

8.8.2 Extremal numbers

$\text{ex}(n, F)$ is the max edges in an n -vertex F -free graph. For bipartite F : $\text{ex}(n, F) = o(n^2)$ (Erdős–Stone). For $K_{s,t}$: $\text{ex}(n, K_{s,t}) = O(n^{2-1/s})$ (Kővári–Sós–Turán).

8.9 Spectral graph theory

8.9.1 Spectra of standard graphs

K_n : $n - 1$ (1), -1 ($n - 1$). $K_{m,n}$: $\pm\sqrt{mn}$, 0 ($m + n - 2$). C_n : $2 \cos(2\pi k/n)$. P_n : $2 \cos(\pi k/(n + 1))$.

8.9.2 Hoffman bound for $\alpha(G)$

G is k -regular on n vertices, λ_n is the smallest eigenvalue $\Rightarrow \alpha(G) \leq \frac{-n\lambda_n}{k - \lambda_n}$.

8.6.3 Wilf bound

$\chi(G) \leq 1 + \lambda_1(G)$, where λ_1 is the largest eigenvalue of the adjacency matrix.

8.6.4 Mycielski construction

For any graph G with $\chi(G) = k$, the Mycielski graph $M(G)$ has $\chi(M(G)) = k+1$ and contains no triangles (if G had none). Hence there exist triangle-free graphs with arbitrarily large chromatic number.

8.7 Planar graphs

8.7.1 Euler's formula

For a connected planar graph drawn in the sphere: $n - m + f = 2$ (f = number of faces). Consequence: $m \leq 3n - 6$ for $n \geq 3$ (each face has ≥ 3 edges); equality holds for triangulations (all faces triangles). A triangle-free planar graph satisfies $m \leq 2n - 4$. From $m \leq 3n - 6$ and $2m \geq n\delta$, every planar graph has $\delta \leq 5$.

K_5 and $K_{3,3}$ are non-planar. *Kuratowski*: a graph is planar $\iff$ it contains no subdivision of K_5 or $K_{3,3}$. *Wagner*: a graph is planar $\iff$ it has no K_5 or $K_{3,3}$ minor.

8.7.2 Five-color theorem

Every planar graph is 5-colorable. (4-color theorem: every planar graph is 4-colorable, but the proof is computer-assisted.)

6.1.10 Formula $|HK|$

$H, K \leq G$ finite $\Rightarrow |HK| = |H| \cdot |K| / |H \cap K|$.

6.1.11 Index of intersection

$[G : H \cap K] \leq [G : H] \cdot [G : K]$.

6.1.12 Chinese remainder for cyclic groups

$\gcd(m, n) = 1 \Rightarrow \mathbb{Z}/m \times \mathbb{Z}/n \cong \mathbb{Z}/(mn)$.

6.1.13 $g^2 = e$ everywhere $\Rightarrow G$ is abelian

If $\forall g \in G$ $g^2 = e$, then G is abelian. If G is finite — $G \cong (\mathbb{Z}/2)^k$.

6.2 Group action: orbit–stabilizer, class equation

6.2.1 Orbit–stabilizer theorem

G acts on X , $x \in X$. The map $gG_x \mapsto gx$ is a bijection $G/G_x \rightarrow Gx$. When $|G| < \infty$:

$$|Gx| = [G : G_x] = \frac{|G|}{|G_x|}.$$

6.2.2 Class equation (conjugation)

G acts on itself by conjugation $g \cdot h = ghg^{-1}$.

1. $|[h]| = [G : C_G(h)]$.
2. $|G| = |Z(G)| + \sum_i [G : C_G(g_i)]$ (over non-central classes).
3. The number of conjugates of a subgroup H equals $[G : N_G(H)]$.

6.2.3 Fixed point lemma for p -groups

P is a p -group acting on a finite $X \Rightarrow |X^P| \equiv |X| \pmod{p}$. If $p \nmid |X|$, there is a fixed point.

6.2.4 Center of a non-trivial finite p -group is non-trivial

$|G| = p^n, n \geq 1 \Rightarrow p \mid |Z(G)|$ (in particular $Z(G) \neq \{e\}$).

6.2.5 Burnside's lemma (Cauchy–Frobenius)

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where X^g are the fixed points of g . Pólya variant: $G \leq S_n$, number of orbits on functions $[n] \rightarrow Y$, $|Y| = k$: $\frac{1}{|G|} \sum_g k^{c(g)}$, $c(g)$ is the number of cycles of g .

6.2.6 Cayley's theorem (generalized)

1. G embeds into $S_{|G|}$ (by left translation).
2. If $[G : H] = n$, the action on G/H gives $\varphi : G \rightarrow S_n$ with $\ker \varphi = \bigcap_g gHg^{-1}$. The kernel is a normal subgroup inside H , $[G : \ker \varphi] \mid n!$.

6.3 Conjugacy classes, normalizer, centralizer

6.3.1 Class size divides index of center

$|\text{cl}(g)| \mid [G : Z(G)]$. $|\text{cl}(g)| = 1 \iff g \in Z(G)$.

8.5 Connectivity and orientation

8.5.1 Strong connectivity

A directed graph D is *strongly connected* if for any u, v there is a directed path $u \rightsquigarrow v$. *Strongly connected components (SCCs)* are maximal strongly connected subgraphs; they form a DAG (the *condensation*). Every SCC has at least one cycle (unless it is a single vertex with no loop). A directed graph is strongly connected $\iff$ it has a closed spanning walk.

Robbins' theorem: an undirected graph has a strongly connected orientation $\iff$ it is 2-edge-connected.

8.5.2 Blocks and cut-vertices

A *cut-vertex* (articulation point) is a vertex whose removal increases the number of components. A *block* is a maximal 2-connected subgraph; blocks intersect at cut-vertices forming a tree (block-cut tree). A graph with no cut-vertices is 2-connected $\iff$ any two vertices lie on a common cycle.

8.6 Graph coloring

8.6.1 Lower bounds

$\chi(G) \geq \omega(G)$ (clique number). $\chi(G) \geq \lceil n/\alpha(G) \rceil$ (each color class is independent). For an odd cycle C_{2k+1} : $\chi = 3, \omega = 2, \alpha = k$, so $\chi = n/\alpha$ can be tight.

8.6.2 Greedy bound and Brooks' theorem

$\chi(G) \leq \Delta(G) + 1$ (greedy coloring). *Brooks*: if G is connected and neither a complete graph nor an odd cycle, then $\chi(G) \leq \Delta(G)$.

8.4 Eulerian and Hamiltonian graphs

8.4.1 Eulerian trails

Undirected: Eulerian circuit $\iff$ all degrees even. Eulerian trail $\iff$ exactly 0 or 2 odd degrees. Directed: Eulerian circuit $\iff$ each vertex has equal in- and out-degree. Eulerian trail $\iff$ at most one vertex with out-in = 1, at most one with in-out = 1, others balanced.

8.4.2 Hamiltonicity conditions

Necessary: a Hamiltonian graph is 2-connected and has no cut-vertex. *Sufficient:*

- Dirac: $\delta \geq n/2 \Rightarrow$ Hamiltonian.
- Ore: $\deg u + \deg v \geq n$ for any non-adjacent $u, v \Rightarrow$ Hamiltonian.
- Bondy–Chvátal: if $\deg u + \deg v \geq n$ for non-adjacent u, v , add edge uv . Repeating gives the *Hamiltonian closure*; G is Hamiltonian $\iff$ its closure is K_n .

8.4.3 Tournaments

A *tournament* is a complete oriented graph. Every tournament has a Hamiltonian path. A tournament is *strongly connected* (any vertex reachable from any other) $\iff$ it has a Hamiltonian cycle. Every tournament has a king (a vertex reachable from every other in ≤ 2 steps).

6.3.2 $G/Z(G)$ cyclic $\Rightarrow G$ abelian

If $G/Z(G)$ is cyclic, then G is abelian (and $G/Z(G)$ is trivial).

6.3.3 N/C theorem

$H \leq G \Rightarrow N_G(H)/C_G(H) \hookrightarrow \text{Aut}(H)$. Special case: $G/Z(G) \cong \text{Inn}(G) \leq \text{Aut}(G)$.

6.3.4 Normalizer of a cyclic subgroup

$\text{ord}(g) = n \Rightarrow N_G(\langle g \rangle)/C_G(g) \hookrightarrow (\mathbb{Z}/n)^\times$. Consequence: $[N_G(\langle g \rangle) : C_G(g)] \mid \varphi(n)$.

6.3.5 Class size in a p -group

$|G| = p^n, g \notin Z(G) \Rightarrow |\text{cl}(g)| = p^k, k \geq 1$.

6.4 S_n and A_n : sign, structure, conjugacy

6.4.1 Sign as a homomorphism

The unique $\text{sgn} : S_n \rightarrow \{\pm 1\}$ with $\text{sgn}(\text{transposition}) = -1$. $\ker \text{sgn} = A_n, [S_n : A_n] = 2$.

6.4.2 Cycle type is a complete invariant for conjugacy in S_n

$\sigma, \tau \in S_n$ are conjugate $\Leftrightarrow$ they have the same cycle type. Class size with type $1^{m_1} 2^{m_2} \dots n^{m_n}$: $\frac{n!}{\prod_k m_k! k^{m_k}} \cdot |C_{S_n}(\sigma)| = \prod_k m_k! k^{m_k}$.

6.4.3 Conjugation formula

$$\rho(a_1 \dots a_k) \rho^{-1} = (\rho(a_1) \dots \rho(a_k)).$$

6.4.4 Sign via cycle structure

$\text{sgn}(\sigma) = (-1)^{n-c(\sigma)} = (-1)^{\sum(\lambda_i-1)}$, where $c(\sigma)$ is the number of cycles. A k -cycle has sign $(-1)^{k-1}$.

6.4.5 Generators of S_n

S_n is generated by: all transpositions; adjacent transpositions; the pair $(1\ 2)$ and $(1\ 2 \dots n)$.

6.4.6 Generators of A_n

A_n for $n \geq 3$ is generated by 3-cycles $(1\ 2\ k)$, $k = 3, \dots, n$.

6.4.7 Splitting of S_n -classes in A_n

A class of $\sigma \in A_n$ in S_n splits into two classes in $A_n \Leftrightarrow$ the cycle type of σ consists of pairwise distinct odd lengths.

6.4.8 Simplicity of A_n ($n \geq 5$)

A group $G \neq \{e\}$ is *simple* if it has no non-trivial normal subgroups. A_n is simple for $n \geq 5$.

6.4.9 Normal subgroups of S_n

$n = 1, 2$: $\{e\}, S_n$. $n = 3$: $\{e\}, A_3, S_3$. $n = 4$: $\{e\}, V_4, A_4, S_4$.
 $n \geq 5$: $\{e\}, A_n, S_n$.

8.3 Spanning subgraphs and matchings

8.3.1 Hall's theorem

A bipartite graph $G = (X \cup Y, E)$ has a matching saturating $X \iff |N(S)| \geq |S|$ for all $S \subseteq X$.

8.3.2 Menger's theorem

The minimum number of vertices separating non-adjacent a, b equals the maximum number of internally vertex-disjoint a - b paths. Edge version: min edge cut size = max number of edge-disjoint paths. Consequence: G is k -connected $\iff$ any two vertices are linked by k internally vertex-disjoint paths.

8.3.3 Bipartite graphs and König's theorem

G is bipartite $\iff$ it has no odd cycles. For bipartite G : $\nu(G) = \tau(G)$ (König), where ν is max matching and τ is min vertex cover. *Gallai's theorem*: for any graph without isolated vertices, $\nu(G) + \beta(G) = n$, where $\beta(G)$ is the size of a minimum edge cover. Consequence: $\tau(G) + \beta(G) = n$ for bipartite graphs.

8.3.4 Independent sets and covers

For any graph: a set S is independent $\iff V \setminus S$ is a vertex cover, hence $\alpha(G) + \tau(G) = n$. Together with König, for bipartite graphs $\alpha(G) = n - \nu(G)$. Gallai gives $\beta(G) = \nu(G)$ for bipartite graphs (since $\tau = \nu$). Simple bound: $\alpha(G) \geq n/(\Delta + 1)$ (greedy: pick a vertex, delete it and its neighbours, repeat).

A graph is *k-connected* ($\kappa(G) \geq k$) if removing $< k$ vertices leaves it connected; *k-edge-connected* ($\lambda(G) \geq k$) similarly.

8.1.2 Distance and diameter bounds

$\text{diam}(G) \leq n - 1$ (equality for a path). If $\delta \geq 2$ then $\text{diam}(G) \leq n/2$. For a tree: $\text{diam} \geq 2\text{rad} - 1$ or 2rad depending on parity. $\text{rad}(G) \leq \text{diam}(G) \leq 2\text{rad}(G)$.

8.1.3 Degree sequences

Handshaking lemma: $\sum \deg v = 2m$, hence the number of odd-degree vertices is even. A sequence is *graphic* (realizable as a simple graph) iff it satisfies the Erdős–Gallai inequalities.

8.2 Trees

8.2.1 Tree characterizations

For G on n vertices the following are equivalent: (1) tree, (2) connected + $m = n - 1$, (3) acyclic + $m = n - 1$, (4) unique path between any two vertices. Every tree with $n \geq 2$ has at least two leaves.

8.2.2 Cayley's formula and Prüfer code

Number of labeled trees on n vertices = n^{n-2} . Corollary: number of spanning trees of K_n is n^{n-2} .

8.2.3 Kirchhoff's theorem (matrix-tree)

$t(G) = \det \tilde{L}_{ii} = \frac{1}{n} \lambda_2 \cdots \lambda_n$, where $L = D - A$ is the Laplacian, $\lambda_1 \geq \cdots \geq \lambda_n = 0$ are its eigenvalues.

6.4.10 Commutator

The *commutator subgroup* $[G, G] = \langle g^{-1}h^{-1}gh : g, h \in G \rangle$ is the smallest normal subgroup with abelian quotient. $[S_n, S_n] = A_n$ ($n \geq 2$). $[A_n, A_n] = A_n$ for $n \geq 5$ (perfect). $[A_4, A_4] = V_4$.

6.4.11 Solvability of S_n, A_n

A group G is *solvable* if its derived series $G^{(0)} = G, G^{(k+1)} = [G^{(k)}, G^{(k)}]$ reaches $\{e\}$ finitely. S_n and A_n are solvable $\Leftrightarrow n \leq 4$.

6.5 Sylow theorems and p -groups

6.5.1 Sylow theorems

$|G| = p^a m, \gcd(p, m) = 1$.

- *I (existence)*. For each $0 \leq k \leq a$ there is a subgroup of order p^k . A subgroup of order p^a is a Sylow p -subgroup.
- *II (conjugacy)*. All Sylow p -subgroups are conjugate. Any p -subgroup lies in some Sylow p -subgroup.
- *III (counting)*. $n_p \equiv 1 \pmod{p}$, $n_p \mid m$, $n_p = [G : N_G(P)]$.

6.5.2 Frattini argument

$N \trianglelefteq G, P \in \text{Syl}_p(N) \Rightarrow G = N \cdot N_G(P)$.

6.5.3 p -groups are nilpotent

A group G is *nilpotent* if its upper central series $\{e\} = Z_0 \leq Z_1 \leq \cdots \leq Z_k = G$ reaches G in finitely many steps ($Z_{i+1}/Z_i = Z(G/Z_i)$). For $|G| = p^n$ this holds in at most n steps. Consequence: in a p -group every proper H is strictly smaller than $N_G(H)$.

6.5.4 Maximal subgroup of a p -group

$|G| = p^n$, M is maximal $\Rightarrow M \trianglelefteq G$ and $[G : M] = p$.

6.5.5 Frattini subgroup

$\Phi(G) = \bigcap M$ (intersection of all maximal subgroups). For a p -group: $\Phi(G) = [G, G] \cdot G^p$. $G/\Phi(G) \cong (\mathbb{F}_p)^d$, d is the minimal number of generators.

6.5.6 Chain of normal subgroups in a p -group

$|G| = p^n \Rightarrow \exists \{e\} = N_0 \subsetneq \cdots \subsetneq N_n = G$ with $|N_k| = p^k$ and $N_k \trianglelefteq G$.

6.6 Finite abelian groups

6.6.1 Structure theorem

Any finite abelian G :

- *Invariant factors*: $G \cong \mathbb{Z}/d_1 \oplus \cdots \oplus \mathbb{Z}/d_k$, $1 < d_1 \mid d_2 \mid \cdots \mid d_k$.
- *Elementary divisors*: $G \cong \bigoplus \mathbb{Z}/p_i^{a_i}$ — uniquely determined up to permutation.

6.6.2 p -Sylow decomposition

G abelian $\Rightarrow G = \bigoplus_{p \mid |G|} G_p$, where G_p is the p -Sylow subgroup.

6.6.3 $G[n]$ in an abelian group

$G \cong \bigoplus \mathbb{Z}/p_i^{a_i} \Rightarrow |G[n]| = \prod_i \gcd(n, p_i^{a_i})$.

Chapter 8

Graph Theory

Notation. $G = (V, E)$ with $n = |V|$, $m = |E|$. $\kappa(G)$ — vertex connectivity; $\lambda(G)$ — edge connectivity. $\delta(G)$ — minimum degree; $\Delta(G)$ — maximum degree. $\chi(G)$ — chromatic number; $\alpha(G)$ — independence number. $\omega(G)$ — clique number; $\nu(G)$ — matching number; $\tau(G)$ — vertex cover number; $\beta(G)$ — edge cover number. $\text{diam}(G)$ — diameter; $\text{rad}(G)$ — radius. $N(v)$ — neighbourhood of v ; $N(S) = \bigcup_{v \in S} N(v)$. $\overline{G}$ — complement; $G[U]$ — induced subgraph. G is k -regular if $\deg(v) = k$ for all v .

8.1 Basic definitions and facts

8.1.1 Walks, paths, cycles, connectivity

A *walk* is a sequence of vertices with consecutive ones adjacent; a *path* is a walk with no repeated vertices; a *cycle* is a closed path. G is *connected* if any two vertices are linked by a path; *components* are maximal connected subgraphs. The *distance* $d(u, v)$ is the length of a shortest path; *diameter* $\text{diam}(G) = \max_{u, v} d(u, v)$; *radius* $\text{rad}(G) = \min_u \max_v d(u, v)$. The center consists of vertices with $\max_v d(u, v) = \text{rad}(G)$; for a tree it is a single vertex or two adjacent vertices.

6.6.4 Converse of Lagrange for abelian groups

G abelian, $d \mid |G| \Rightarrow \exists H \leq G$ of order d .

6.6.5 Duality of characters

G abelian. A *character* is a homomorphism $\chi : G \rightarrow \mathbb{C}^\times$. $\widehat{G} = \text{Hom}(G, \mathbb{C}^\times) \cong G$. Orthogonality: $\sum_g \chi(g) \overline{\psi(g)} = |G| \cdot [\chi = \psi]$; $\sum_\chi \chi(g) \overline{\chi(h)} = |G| \cdot [g = h]$.

6.6.6 Smith normal form

$M \in \mathbb{Z}^{m \times n}$ can be reduced to $\text{diag}(d_1, \dots, d_r, 0, \dots)$ with $d_1 \mid d_2 \mid \dots \mid d_r$ by multiplication by $GL_m(\mathbb{Z}), GL_n(\mathbb{Z})$. Application: if $G \cong \mathbb{Z}^n / M\mathbb{Z}^m$, then $G \cong \mathbb{Z}^{n-r} \oplus \bigoplus \mathbb{Z}/d_i$ (see also 3.14.1).

6.7 Basic methods and techniques

6.7.1 Semidirect products

Definition. $G = N \rtimes H$ where $N \trianglelefteq G$, $H \leq G$, $NH = G$, $N \cap H = \{e\}$. Multiplication is determined by $\varphi : H \rightarrow \text{Aut}(N)$:

$$(n_1, h_1)(n_2, h_2) = (n_1 \varphi_{h_1}(n_2), h_1 h_2).$$

Examples. $D_n \cong C_n \rtimes C_2$ (inversion); classification of groups of order pq (6.1.8) gives $\mathbb{Z}/q \rtimes \mathbb{Z}/p$ when $p \mid q-1$.

Recognition. If $N \trianglelefteq G$, $G = NH$, $N \cap H = \{e\}$, then $G \cong N \rtimes H$.

6.7.2 Common techniques

- *Sylow counting.* $n_p \equiv 1 \pmod{p}$, $n_p \mid m$ often determines n_p . $n_p = 1 \Rightarrow$ the Sylow subgroup is normal $\Rightarrow$ the group is not simple.

- *Element counting.* If n_p is large, elements of order p occupy much space, leaving little for others — often $n_q = 1$.
- *Class equation.* $|G| = |Z(G)| + \sum [G : C_G(g)]$ is the foundation of p -group induction.
- *Double counting.* The kernel of Burnside's lemma; counting pairs (g, x) with $gx = x$.
- *Action on cosets.* Turns a subgroup H into a homomorphism $G \rightarrow S_{[G:H]}$. The kernel is $\text{core}_G(H)$, a normal subgroup inside H .
- *N/C theorem.* $N_G(H)/C_G(H) \hookrightarrow \text{Aut}(H)$. Restricts the normalizer via automorphisms.
- *Reduction to p -parts.* In the abelian case the problem splits into independent p -components.

7.9.2 Bonferroni inequalities

Even truncation $m = 2r$: $|\bigcup A_i| \geq \sum_{k=1}^{2r} (-1)^{k+1} S_k$. Odd $m = 2r + 1$: $|\bigcup A_i| \leq \sum_{k=1}^{2r+1} (-1)^{k+1} S_k$.

7.9.3 Möbius inversion

On a locally finite poset: $g(y) = \sum_{x \leq y} f(x) \iff f(y) = \sum_{x \leq y} \mu(x, y)g(x)$.

For the Boolean lattice: $\mu(I, J) = (-1)^{|J \setminus I|}$. For the divisor lattice: $\mu(d, n) = \mu(n/d)$.

7.9.4 Euler's totient φ

$\varphi(n) = n \prod_{p|n} (1 - 1/p) = \sum_{d|n} \mu(d)n/d$. Möbius inversion: $\sum_{d|n} \varphi(d) = n$.

7.9.5 Chromatic polynomial

$P(G, x) = \sum_{S \subseteq E} (-1)^{|S|} x^{c(V, S)}$ (inclusion-exclusion), where $c(V, S)$ is components of (V, S) . Recurrence: $P(G, x) = P(G \setminus e, x) - P(G/e, x)$.

7.8 Combinatorial geometry

7.8.1 Helly's theorem

Convex sets $K_1, \dots, K_m \subseteq \mathbb{R}^d$, $m \geq d + 1$. If any $d + 1$ of them have a common point, then $\bigcap_{i=1}^m K_i \neq \emptyset$.

When to use. Problems about coverings, intersections of convex figures, point arrangements.

7.8.2 Sylvester–Gallai theorem

For a finite set of points in the plane not all on one line, there exists a line containing exactly two of them.

Idea. Consider the pair (point, line) with minimal distance; if the line has ≥ 3 points, another pair with smaller distance is found.

7.8.3 Minkowski's convex body theorem

$K \subset \mathbb{R}^n$ is a convex centrally symmetric ($K = -K$) body with $\text{vol}(K) > 2^n$. Then K contains a non-zero integer point $z \in \mathbb{Z}^n \setminus \{0\}$.

When to use. Diophantine approximation, existence of solutions to systems of inequalities, quadratic forms.

7.9 Inclusion–exclusion and Möbius inversion

7.9.1 Inclusion–exclusion principle (PIE)

$|\bigcap \overline{A_i}| = \sum_{k=0}^n (-1)^k S_k$, where $S_k = \sum_{|I|=k} |\bigcap_{i \in I} A_i|$. Works for measures/probabilities.

Chapter 7

Combinatorics

Notation. $[n] = \{1, \dots, n\}$; $\binom{[n]}{k}$ — k -subsets of $[n]$. $2^{[n]}$ — power set; $\binom{n}{k}$ — binomial coefficient. $|X|$ — cardinality; $\mathbb{E}[X]$ — expectation; $\text{Var}(X)$ — variance. $\mathcal{F}$ — family of sets. $A + B = \{a + b : a \in A, b \in B\}$ — sumset; $nA = A + \dots + A$ (n times).

7.1 Pigeonhole principle and extremal principle

7.1.1 Strengthened pigeonhole principle

n objects into k boxes $\Rightarrow$ some box contains $\geq \lceil n/k \rceil$ objects.
Averaging: $\max a_i \geq S/n$, $\min a_i \leq S/n$.

7.1.2 Extremal principle

On a finite set of configurations there exist minimizing and maximizing a given function. Choose a configuration with minimal/maximal value in the “bad” class, locally vary $\rightarrow$ contradiction.

7.1.3 Erdős–Szekeres theorem

A sequence of $N \geq (r-1)(s-1) + 1$ distinct numbers contains an increasing subsequence of length r or a decreasing subsequence of length s . The bound is sharp.

7.1.4 Dilworth and Mirsky theorems

A *poset* is a set with a partial order $\leq$. A *chain* is a totally ordered subset; an *antichain* is a set of pairwise incomparable elements. In a finite poset: the minimum number of chains covering P equals the width $w(P)$ (Dilworth). The minimum number of antichains covering P equals the height $h(P)$ (Mirsky). Consequence: $|P| \geq ab + 1 \Rightarrow$ chain of length $a + 1$ or antichain of size $b + 1$.

7.1.5 Erdős–Ko–Rado theorem

$\mathcal{F} \subseteq \binom{[n]}{k}$ is an intersecting family, $n \geq 2k \Rightarrow |\mathcal{F}| \leq \binom{n-1}{k-1}$. Equality holds for a star (all k -subsets containing a fixed element).

7.1.6 Ramsey theorem (binomial bound)

$R(s, t) \leq R(s-1, t) + R(s, t-1)$, hence $R(s, t) \leq \binom{s+t-2}{s-1}$ and $R(s, s) \leq 4^{s-1}$.

7.2 Probabilistic method

7.2.1 Linearity of expectation

$\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$ for any (dependent) integrable X_i .

7.2.2 First moment method

$X \geq 0$, $\mathbb{E}[X] < 1 \Rightarrow P(X = 0) > 0$ (existence).

7.6.5 Lagrange inversion formula

$w = x\phi(w)$, $\phi(0) \neq 0 \Rightarrow [x^n]H(w) = \frac{1}{n}[t^{n-1}]H'(t)\phi(t)^n$.

7.7 Recurrence relations and closed forms

7.7.1 Closed form of linear recurrences

$a_n = \sum c_i a_{n-i}$, roots λ_i of multiplicity $m_i \Rightarrow a_n = \sum P_i(n)\lambda_i^n$, $\deg P_i \leq m_i - 1$. Non-homogeneous case: particular solution $n^s R(n)\mu^n$ with s equal to the multiplicity of μ as a characteristic root.

7.7.2 Companion matrix and powers

$\mathbf{v}_n = C\mathbf{v}_{n-1}$, C^n can be computed in $O(\log n)$ multiplications.

7.7.3 Derangements

$D_n = n! \sum_{k=0}^n (-1)^k / k!$, $D_n = nD_{n-1} + (-1)^n$. EGF: $e^{-x}/(1-x)$.

7.7.4 Stirling numbers of the second kind

$S(n, k) = kS(n-1, k) + S(n-1, k-1)$. $S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n$. EGF: $(e^x - 1)^k / k!$.

7.7.5 Fibonacci identities

$F_{n-1}F_{n+1} - F_n^2 = (-1)^n$, $F_{m+n} = F_m F_{n+1} + F_{m-1} F_n$. Consequence: $\gcd(F_m, F_n) = F_{\gcd(m, n)}$.

7.5.4 Plünnecke–Ruzsa inequality

If $|A + A| \leq K|A|$, then $|nA - mA| \leq K^{n+m}|A|$, where $nA = A + \dots + A$ (n times).

When to use. Bounds on iterated sumsets under small doubling.

7.5.5 Erdős–Ginzburg–Ziv (EGZ)

Among any $2n - 1$ integers there exist n whose sum is divisible by n .

Generalization (Kemnitz). Among $2n - 1$ elements of any abelian group of order n there are n with sum 0 (see also 5.3.9).

7.6 Generating functions (OGF and EGF)

7.6.1 Basic operations

OGF: $A(x) = \sum a_n x^n$: AB gives convolution, xA gives shift, $A/(1 - x)$ gives partial sums. EGF: $\hat{A}(x) = \sum a_n x^n/n!$: product gives binomial convolution.

7.6.2 Roots of unity filter

$\sum_{n \equiv r \pmod{m}} a_n x^n = \frac{1}{m} \sum_{j=0}^{m-1} \omega^{-jr} A(\omega^j x)$ with $\omega = e^{2\pi i/m}$.

7.6.3 Linear recurrences with constant coefficients

$\sum c_i a_{n+i} = 0 \Rightarrow A(x) = P(x)/Q(x)$, where $Q(x) = c_k + \dots + c_0 x^k = \prod (1 - \alpha_j x)^{m_j}$. Then $a_n = \sum Q_j(n) \alpha_j^n$.

7.6.4 Catalan numbers

$C_{n+1} = \sum C_k C_{n-k}$, $C_0 = 1 \Rightarrow C(x) = 1 + xC(x)^2$, $C_n = \frac{1}{n+1} \binom{2n}{n}$.

7.2.3 Alteration method

A random construction C is almost good. Remove defects. Obtain a pure construction of size $\geq \mathbb{E}[|C|] - \mathbb{E}[Y]$, where Y is the number of defects.

7.2.4 Lovász Local Lemma (LLL)

Events A_i , $P(A_i) \leq p$, each depends on at most d others, $ep(d + 1) \leq 1 \Rightarrow P(\bigcap A_i) > 0$.

7.2.5 Second moment method

$P(X = 0) \leq \frac{\text{Var}(X)}{(\mathbb{E}[X])^2}$. If $\text{Var}(X) = o((\mathbb{E}[X])^2)$, then $X > 0$ with probability $\rightarrow 1$.

7.2.6 Random lower bound for Ramsey numbers

$R(k, k) > 2^{k/2}$ for $k \geq 3$. Improved by alteration: $\frac{k}{e\sqrt{2}} 2^{k/2}$.

7.2.7 Chernoff–Hoeffding inequality

$X_i \in [a_i, b_i]$ independent, $S = \sum X_i$: $P(|S - \mathbb{E}[S]| \geq t) \leq 2 \exp(-2t^2 / \sum (b_i - a_i)^2)$.

7.3 Bijective arguments and double counting

7.3.1 Double counting (incidence matrix)

For $R \subseteq X \times Y$: $|R| = \sum_x |\{y\}| = \sum_y |\{x\}|$. Consequences: $\sum \deg v = 2|E|$, Vandermonde's identity, hockey-stick identity.

7.3.2 LYM inequality and Sperner's theorem

$\mathcal{F} \subseteq 2^{[n]}$ is an antichain $\Rightarrow \sum_{F \in \mathcal{F}} 1/\binom{n}{|F|} \leq 1$. Consequence:
 $|\mathcal{F}| \leq \binom{n}{\lfloor n/2 \rfloor}$.

7.3.3 Burnside's lemma

A finite group G acts on $X \Rightarrow$ number of orbits $= \frac{1}{|G|} \sum_{g \in G} |X^g|$.

7.3.4 Hall's theorem (on matchings)

$A_1, \dots, A_n$ have an SDR $\iff$ for all $S \subseteq [n]$: $|\bigcup_{i \in S} A_i| \geq |S|$.

7.3.5 Cayley's formula and Prüfer code

Number of labeled trees on n vertices $= n^{n-2}$.

7.4 Linear algebra methods

7.4.1 Dimensionality argument

$v_1, \dots, v_m$ are linearly independent in V of dimension $n \Rightarrow m \leq n$.

7.4.2 Diagonal independence criterion

If $f_i(x_j) \neq 0$ at $i = j$ and $= 0$ at $i \neq j$, then f_i are linearly independent.

7.4.3 Oddtown

$A_i \subseteq [n]$, $|A_i|$ odd, $|A_i \cap A_j|$ even for $i \neq j \Rightarrow m \leq n$.

7.4.4 Eventown

$|A_i|$ and $|A_i \cap A_j|$ even for all $i, j \Rightarrow m \leq 2^{\lfloor n/2 \rfloor}$.

7.4.5 Fisher's inequality

$A_i \subseteq [n]$, $|A_i \cap A_j| = \lambda \geq 1$ for $i \neq j \Rightarrow m \leq n$.

7.4.6 Combinatorial Nullstellensatz (Alon)

$f \in \mathbb{F}[x_1, \dots, x_n]$ of degree $d = \sum t_i$, coefficient of $\prod x_i^{t_i} \neq 0$.
 $S_i \subseteq \mathbb{F}$, $|S_i| > t_i \Rightarrow \exists s \in \prod S_i$ with $f(s) \neq 0$ (see also 5.3.8).

7.5 Additive combinatorics

7.5.1 Cauchy–Davenport theorem

For non-empty $A, B \subseteq \mathbb{Z}/p\mathbb{Z}$ (p prime):

$$|A + B| \geq \min(p, |A| + |B| - 1).$$

When to use. Lower bounds on sumsets, existence of solutions $a + b = c$ in subsets of $\mathbb{F}_p$.

Idea. Induction on $|A|$: shift and delete intersection.

7.5.2 Ruzsa triangle inequality

For non-empty $A, B, C \subseteq G$ (abelian group):

$$|A + C| \cdot |B| \leq |A + B| \cdot |B + C|.$$

Corollary. $|A - A| \leq (|A + A|/|A|)^2 |A|$.

7.5.3 Petridis's lemma

For $A, B \subseteq G$ with $|A + B| \leq K|A|$ there exists non-empty $X \subseteq A$ with $|X + B'| \leq K|X|$ for any $B' \subseteq B$, where B' is an arbitrary subset.

